

PUBP 233
INTRODUCTION TO DEMOGRAPHY
(POPULATION AND SOCIETY)

**Week 9: FERTILITY- Adding
people**

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FERTILITY

Adding people

FERTILITY

Fertility: bringing people into the population; determines Population size and structure

- ▢ **Fecundity:** Ability to reproduce; Biological component of fertility ;
- ▢ **Sterility:** refers to permanent inability to conceive (can be for Men or Women)

Fertility : “Risk” of producing live births

Two sex problem, and ‘repeatable event’

Fertility- depends on **biological parameters, social norms and behavior** and **calculus of conscious choice** . An area where **sociological explanations** flood into the more precise measures that demography offers.

Questions:

- ▢ Why do women and couples decide to have more or less children ?
- ▢ How governments have sometimes attempted to influence those decisions.

FACTORS INFLUENCING FERTILITY

Socio-cultural:

- Mothers education
- **Institution of marriage**
- Child bearing and rearing decisions
- Contraception practices
- **Religion**

Economic realities:

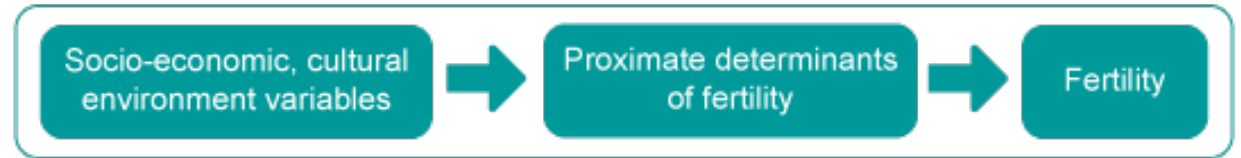
Income levels; support in older ages

Government Policies and Law:

Population policies, Abortion laws

Biological and health related factors:

Child Mortality, HIV/STDs



Source: Bongaarts and Potter (1983)

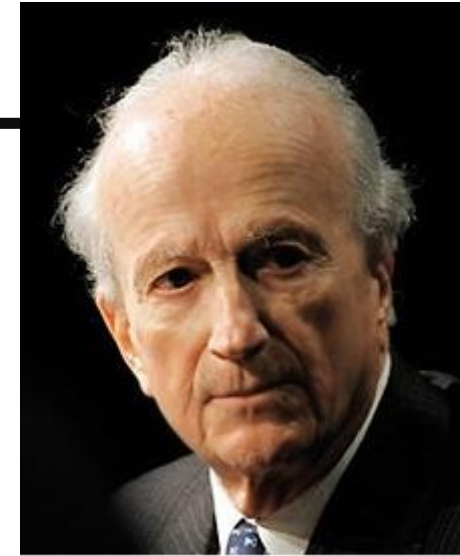
http://papp.iussp.org/sessions/papp101_s05/PAPP101_s05_030_010.html

ECONOMICS OF FERTIL

Gary Becker: Nobel Laureate-Economics (1992) – applied economic theory to a host of traditionally sociological concerns:

- ▮ **Fertility:** Value of children (eg: “*An Economic Analysis of Fertility*”- **quality and quantity** of children)
- ▮ **Household economics:** Behavior of the family (or household), including distribution of work and allocation of time in the family
- ▮ **Marriage, Divorce**, Racial discrimination...
- ▮ Introduced the concept of **Human Capital**
- ▮ **“Rotten kid theorem”**

http://rightweb.irc-online.org/profile/becker_gary/



Gary Becker (1930-2014)
An Economic Analysis of Fertility

GARY S. BECKER

COLUMBIA UNIVERSITY

AND

NATIONAL BUREAU OF ECONOMIC RESEARCH

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THE inability of demographers to predict western birth rates accurately in the postwar period has had a salutary influence on demographic research. Most predictions had been based either on simple extrapolations of past trends or on extrapolations that adjusted for changes in the age-sex-marital composition of the population. Socio-economic considerations

FERTILITY MEASURES

Crude Birth Rate	CBR
Child Woman Ratio	C/W Ratio
General Fertility Rate	GFR
Age-specific Fertility Rates	ASFRs
The Total Fertility Rate	TFR
Parity Progression Ratios	PPRs

$$\text{Crude Birth Rate} = \frac{\text{births in a stated period}}{\text{mean population over that period}}$$

$$\text{General Fertility Rate} = \frac{\text{births in a stated period}}{\text{mean number of women aged 15 - 49 in the same period}}$$

$$\text{Child/Woman Ratio} = \frac{\text{living children aged 0 - 4}}{\text{women aged 15 - 49}}$$

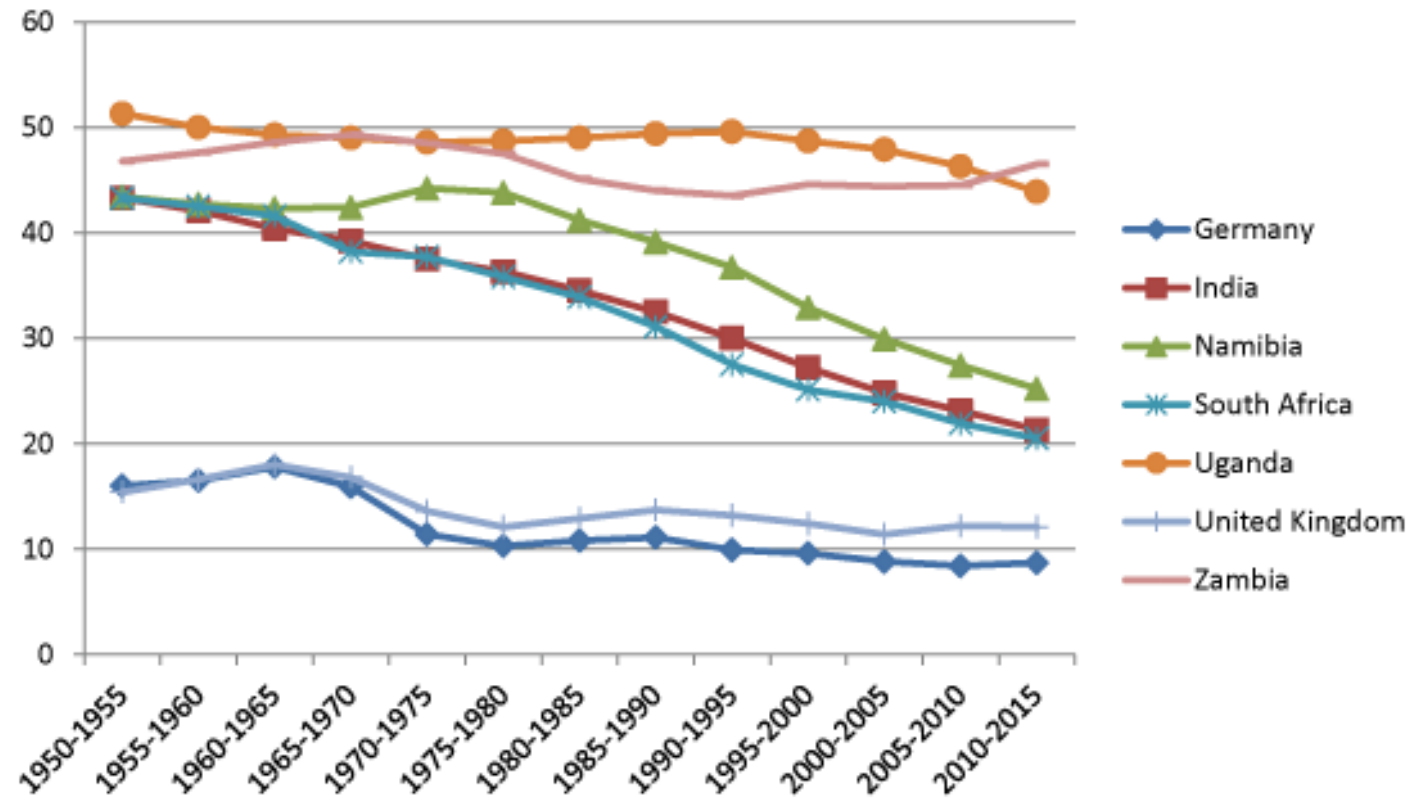
CBR AND GFR

Table 5.1: Comparison of crude birth rates and general fertility rates in selected countries, 1985–90

Country	Total population (thousands)	Number of women aged 15–49 (thousands)	Proportion of women aged 15–49	Annual births (thousands)	Crude birth rate	General fertility rate	rank CBR	rank GFR
Zaire	34,569	7,601	0.220	1,652	0.0478	0.2173	1	2
Kenya	21,747	4,522	0.208	1,002	0.0461	0.2216	2	1
Senegal	6,851	1,530	0.223	312	0.0455	0.2039	3	3
South Africa	35,055	8,401	0.240	1,153	0.0329	0.1372	4	4
India	809,412	191,970	0.237	25,339	0.0313	0.1320	5	5
Tunisia	7,671	1,865	0.243	228	0.0297	0.1223	6	6
Indonesia	175,072	44,325	0.253	4,974	0.0284	0.1122	7	7
Lebanon	2,612	674	0.258	73	0.0279	0.1083	8	8
Argentina	31,436	7,469	0.238	686	0.0218	0.0918	9	9
USA	244,195	64,203	0.263	3,900	0.0160	0.0607	10	10
Australia	16,265	4,243	0.261	246	0.0151	0.0580	11	11
France	55,944	13,801	0.247	772	0.0138	0.0559	12	12
Japan	122,187	31,111	0.255	1,321	0.0108	0.0425	13	13

Data source: United Nations, 1995.

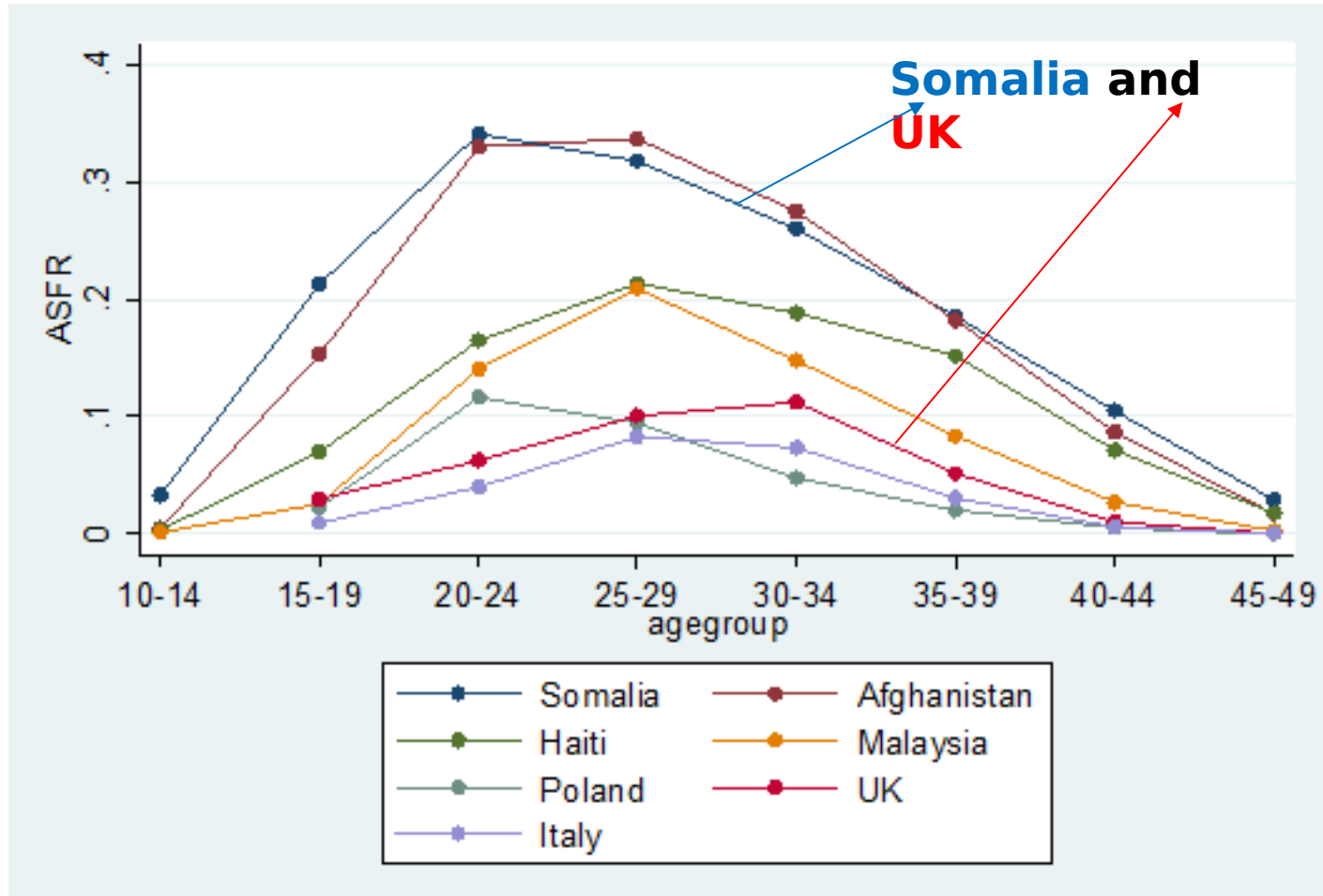
CBR



AGE SPECIFIC FERTILITY RATE (ASFR)

$$ASFR_{x \text{ to } x + n} = \frac{\text{births } x \text{ to } x + n}{\text{females } x \text{ to } x + n} * 1,000$$

ASFR



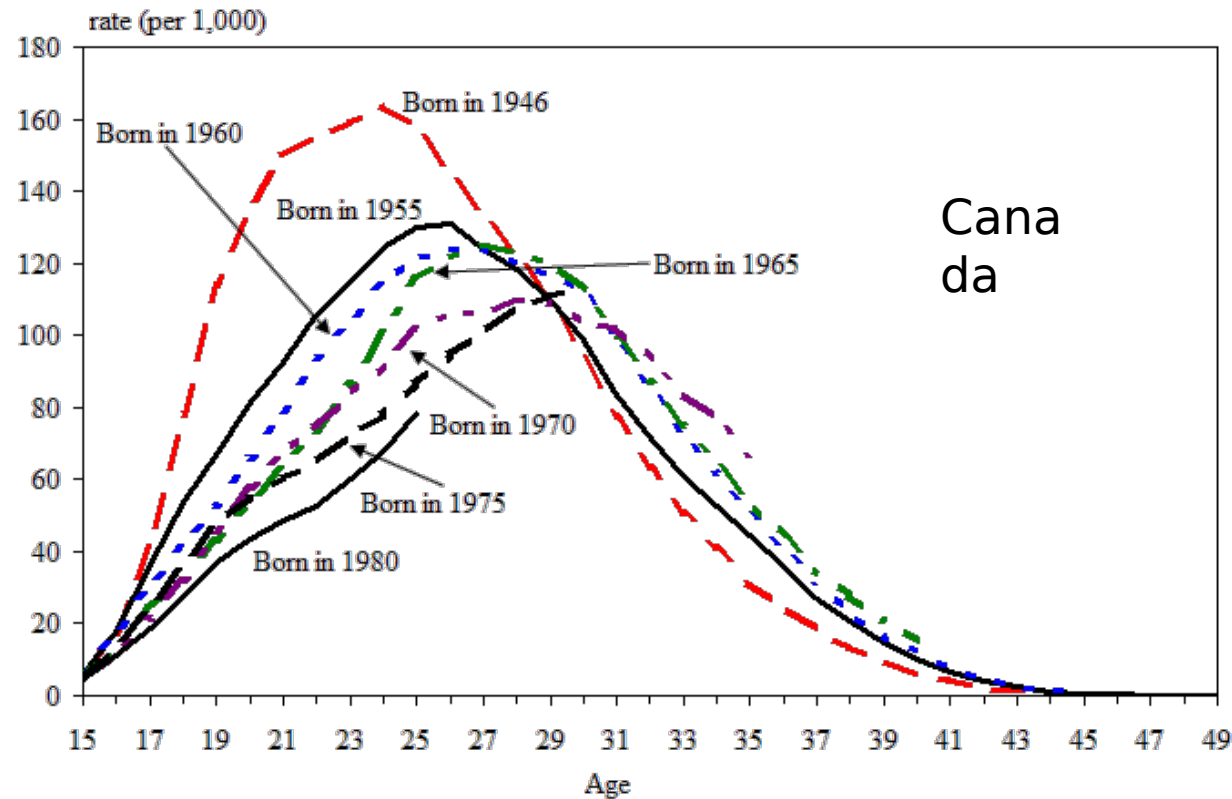
Population Analysis for Policies & Programmes <http://papp.iussp.org/>, Module 101-Session 05

Interactive visual :

http://papp.iussp.org/sessions/papp101_s04/PAPP101_s04_070_030.html

AGE SPECIFIC BIRTH/FERTILITY RATE

$$ASFR_{x \text{ to } x+n} = \frac{\text{births } x \text{ to } x+n}{\text{females } x \text{ to } x+n} * 1,000$$



Canadian statistics,
[http://
www.statcan.gc.ca/](http://www.statcan.gc.ca/)

$f(a)$ = fertility density function

TOTAL FERTILITY RATE

$$TFR = \sum_{x=15}^{45} ASFR_{x \text{ to } x+n} * n \quad n = 5$$

$$TFR = n \sum_{x=\alpha}^{\beta-n} F(x, n)$$

α, β : minimum and maximum age of reproduction

TFR: Average number of children, a women would bear if she **survived through the end of reproductive age span** and experienced at each a particular set of ASFR- $F(x,n)$

$$\text{General Fertility Rate} = \frac{\text{births in a stated period}}{\text{mean number of women aged 15 - 49 in the same period}}$$

$$TFR = \sum_{x=15}^{45} ASFR_{x \text{ to } x+n} * n$$

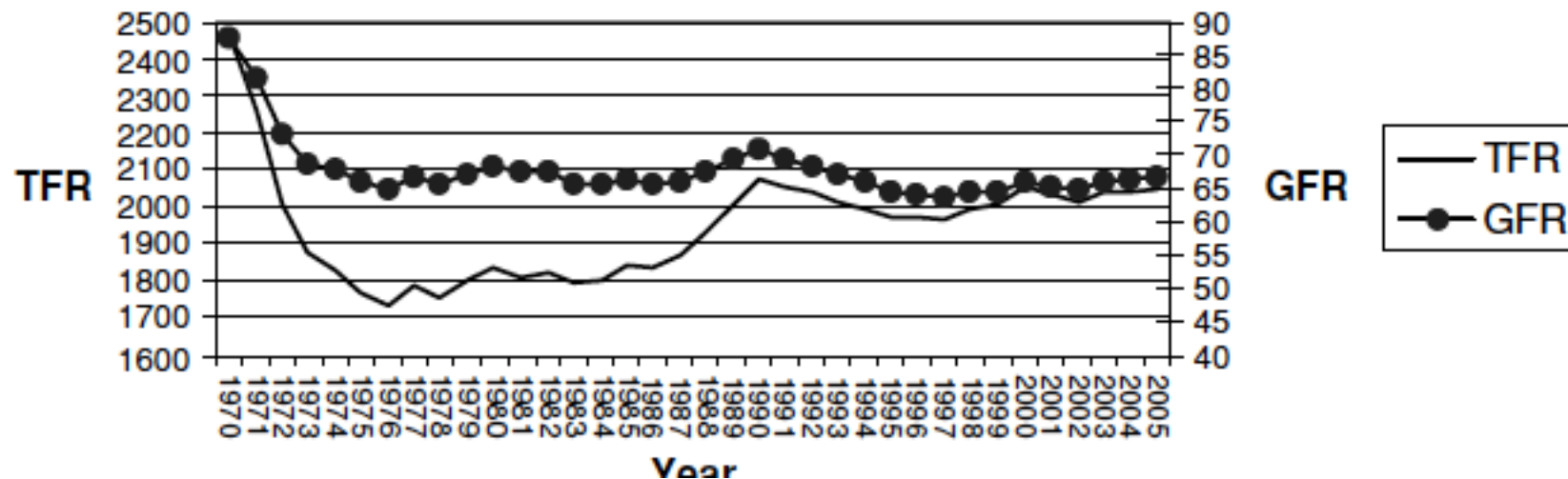
$$ASFR_{x \text{ to } x+n} = \frac{\text{births } x \text{ to } x+n}{\text{females } x \text{ to } x+n} * 1,000$$

$$TFR = CBR * 4.5 * 30$$

$$TFR = GFR * 30$$

(per
thousand)

Approximate
estimates



POPULATION AND SOCIETY

An Introduction to Demography

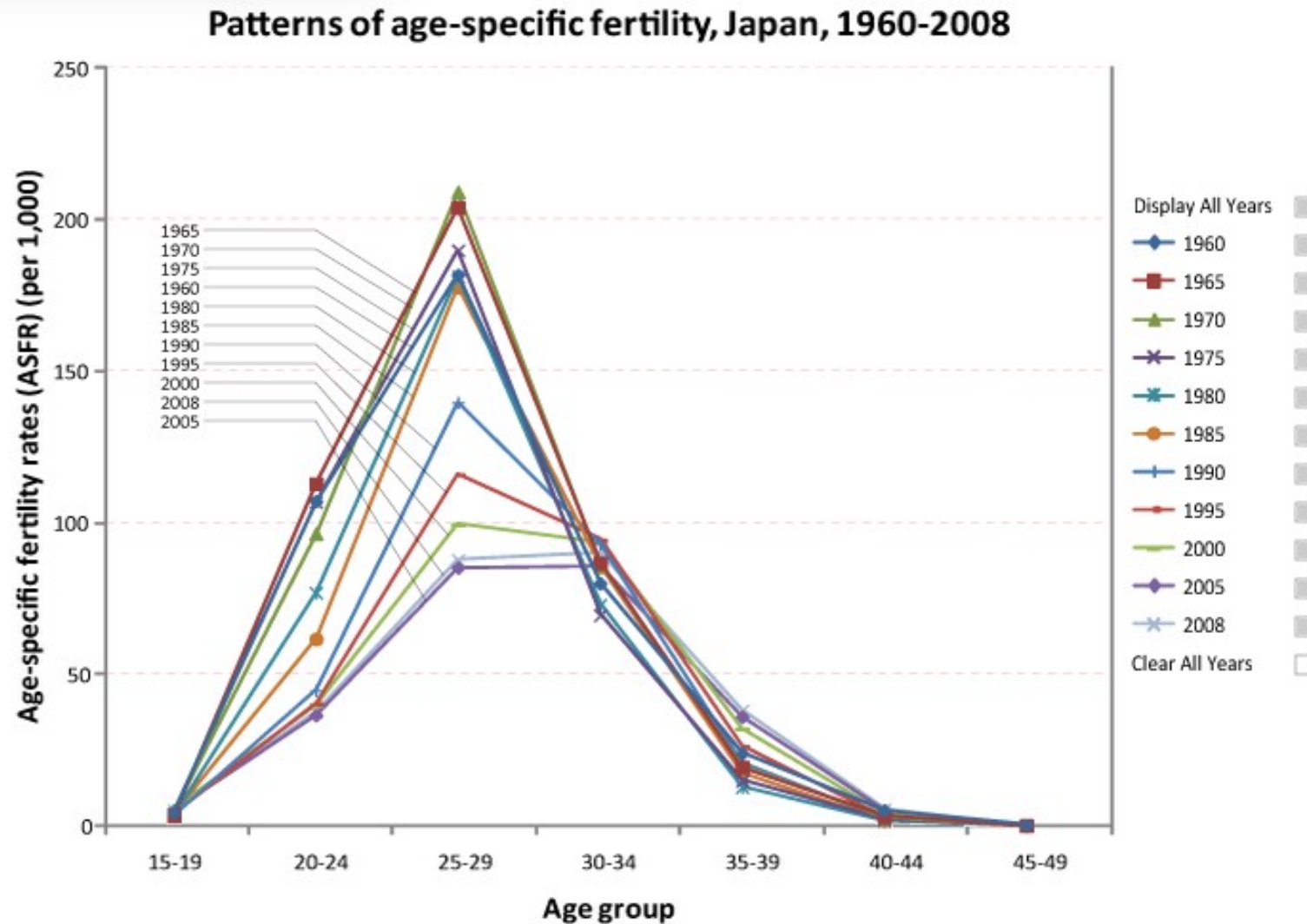
Dudley L. Poston, Jr.

Texas A&M University, College Station, Texas

Leon F. Bouvier

Old Dominion University, Norfolk, Virginia

JAPAN – SHIFT IN ASFR



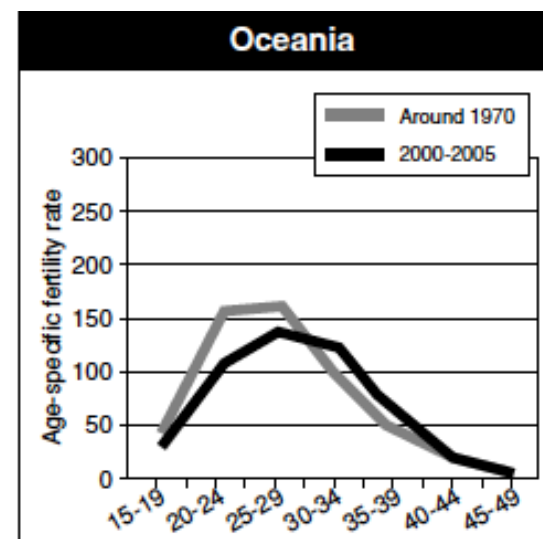
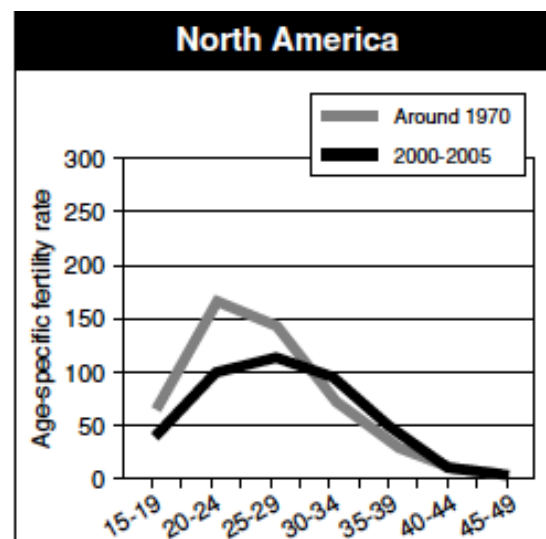
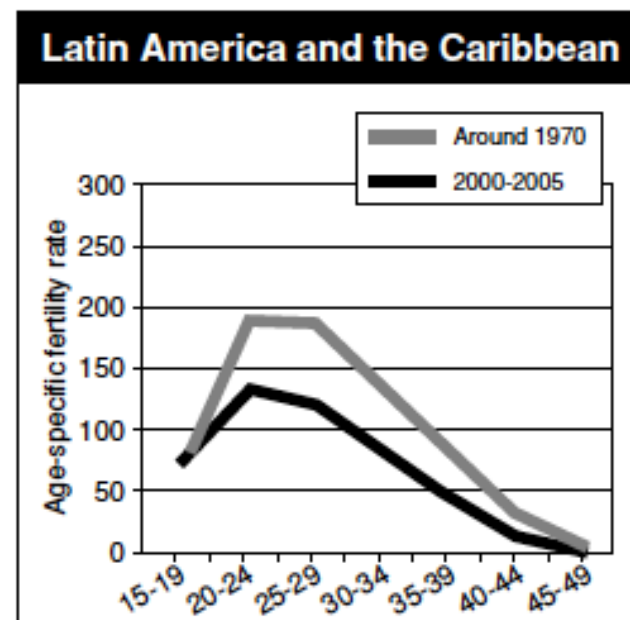
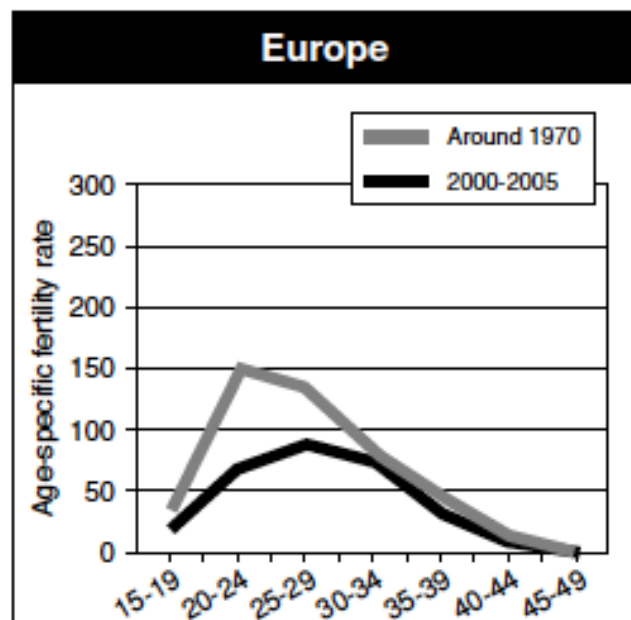
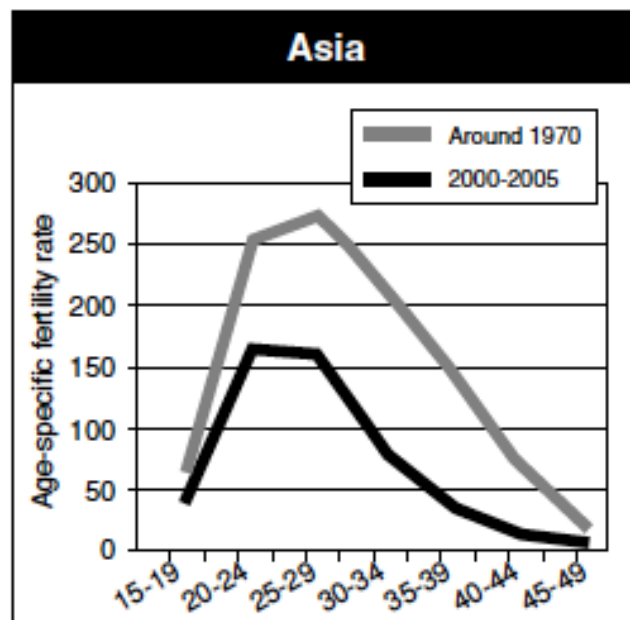
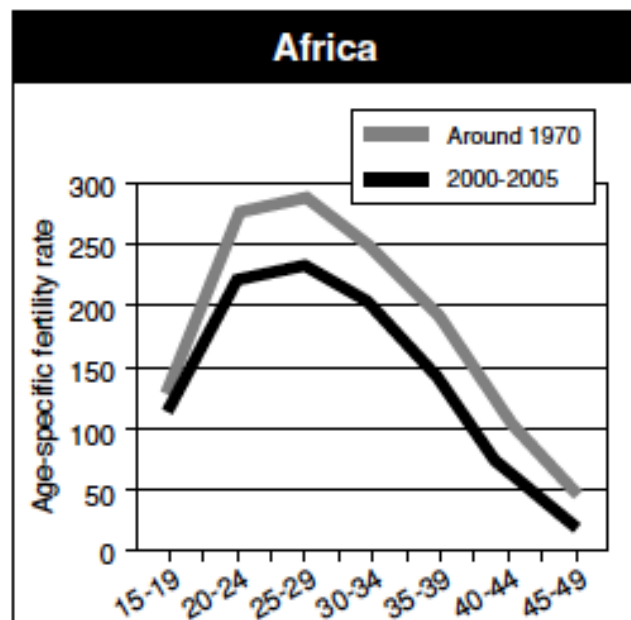


Figure 3.2. Trends in age-specific fertility, regions of the world: 1970 and 2000–2005

TFR CALCULATION

US ,
1992

Age x	${}_5W_x$	${}_5B_x$	${}_5F_x$
10	8,831,206	12,220	0.0014
15	8,324,273	505,415	0.0607
20	9,344,413	1,070,490	0.1146
25	10,047,198	1,179,264	0.1174
30	11,165,144	895,271	0.0802
35	10,619,275	344,644	0.0325
40	9,519,450	55,702	0.0059
45	7,820,172	2,008	0.0003
Sum			0.4128

$$\sum_{x=10,5}^{45} {}_5F_x = .4128$$

$$TFR = 5 \cdot .4128 = 2.064 \text{ children per woman}$$

Data source: National Center for Health Statistics, 1996.

Preston, S., Heuveline, P., & Guillot, M. (2000). *Demography: measuring and modeling population processes*.

Table 3.1. Fertility data and rates for the United States in 2005

Age group	Women in age group (midyear population)	Live births to women in age group	Age-specific fertility rates live births per 1,000 women ASFR	ASFR $\times$ 5
Col. 1	Col. 2	Col. 3	Col. 4	Col. 5
15–19	10,240,239	414,406	40.47	202.3
20–24	10,150,079	1,040,399	102.50	512.5
25–29	9,767,524	1,132,293	115.92	579.6
30–34	9,906,365	952,013	96.10	480.5
35–39	10,427,161	483,401	46.36	231.8
40–44	11,475,863	104,644	9.12	45.6
45–49*	11,372,141	6,546	0.58	2.9
TOTALS	73,339,372	4,133,702		2,055.2 ^a

* Data are for live births to mothers 45+ per 1,000 women 45–49.

^a $TFR = \sum(ASFR \times 5) = 2,055.2$.

Sources: For population data: U.S. Bureau of the Census, 2008; for birth data: Hamilton, Martin, and Ventura, 2007.

GROSS REPRODUCTION RATE AND NET REPRODUCTION RATE

RESTRICTED TO FEMALE BIRTHS

$$GRR = n \sum_{x=\alpha}^{\beta-n} F^f(x, n)$$

$F^f(x, n)$ = Age specific maternity rate (*femle births*)

$$GRR = TFR * \left(\frac{\text{female births}}{\text{births}} \right)$$

$$NRR = \sum_{x=\alpha}^{\beta-n} \frac{L(x, n)}{l_0} F^f(x, n)$$

Considers survivorship of women In births

$$L(x, n) = l(0) * 5$$

$$\frac{L(x, n)}{l(0)} = l(0) * \frac{5}{l(0)} = 5$$

Example: US, 1991; $l_0 = 100,000$

Age x	${}_5W_x$	${}_nB_x^F$	${}_nL_x^F$	${}_nF_x^F$	${}_nF_x^F \cdot {}_nL_x^F$
10	8,620,000	5,816	494,603	0.0007	333.71
15	8,371,000	253,979	493,804	0.0303	14,982.18
20	9,419,000	532,712	492,552	0.0566	27,857.35
25	10,325,000	596,823	491,138	0.0578	28,389.58
30	11,125,000	431,694	489,356	0.0388	18,988.95
35	10,344,000	162,005	486,941	0.0157	7,626.34
40	9,496,000	25,531	483,577	0.0027	1,300.15
45	7,188,000	829	478,475	0.0001	55.18
Sum				0.2026	99,533.45

$$GRR = 5 \cdot 0.2026 = 1.013 \text{ daughter per woman}$$

$$NRR = 99,533.45/100,000 = 0.995 \text{ daughter per woman}$$

GRR, NRR - CONTD

${}_nW_x$ = Mid-year number of women aged x to $x + n$

${}_nB_x^F$ = Number of female births during the year to women aged x to $x + n$

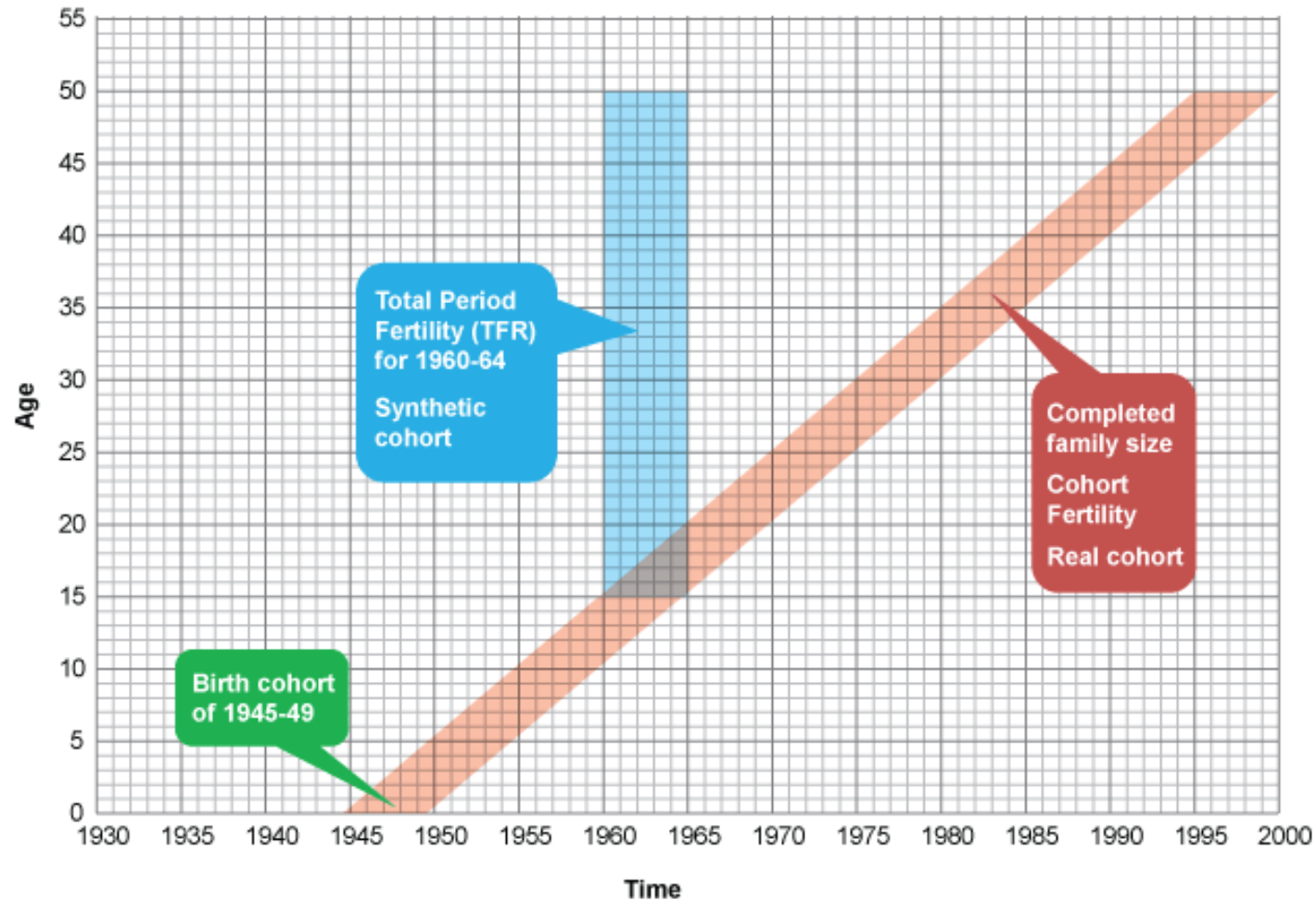
${}_nL_x^F$ = Number of person-years lived between ages x and $x + n$ by a hypothetical cohort of l_0 births in the period life table for females

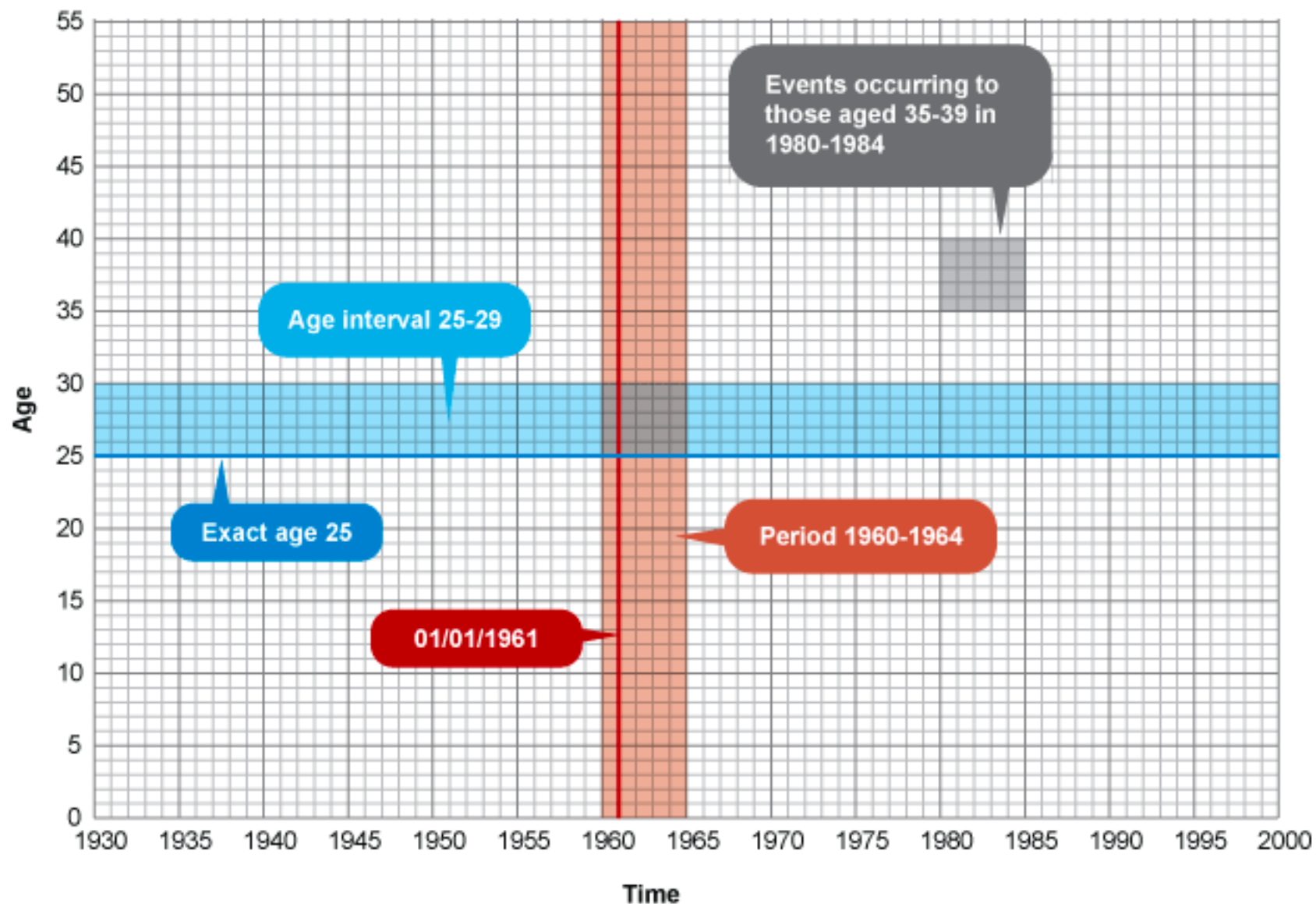
${}_nF_x^F = \frac{{}_nB_x^F}{{}_nW_x}$ = Age-specific maternity rate

$$GRR = n \cdot \sum_{x=\alpha}^{\beta-n} {}_nF_x^F \quad NRR = \frac{1}{l_0} \cdot \sum_{x=\alpha}^{\beta-n} {}_nF_x^F \cdot {}_nL_x^F$$

Preston, S., Heuveline, P., & Guillot, M. (2000). *Demography: measuring and modeling population processes*.

LEXIS DIAGRAM- FERTILITY





Measures of age by calendar time (age-period measures) are represented by the square boxes within the grid. These show demographic events which occur to people within a specific age interval during a specific time interval.

http://papp.iussp.org/sessions/papp101_s02/PAPP101_s02_100_020.html

COHORT AND PERIOD FERTILITY-DIFFERENCE

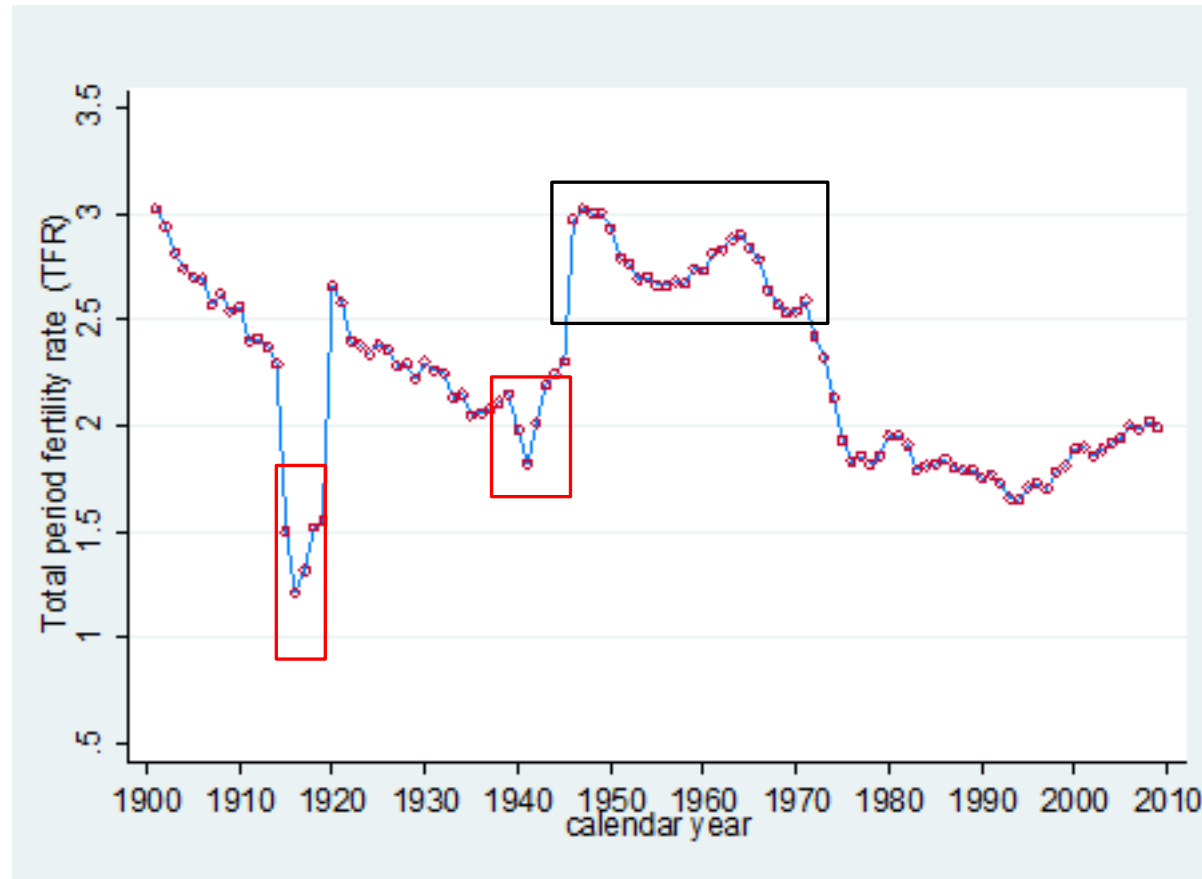
Table 5.3: *Illustration of the distinction between period and cohort total fertility rates*

Age	Age-specific fertility rates								
	Period 1930–4	1935–9	1940–4	1945–9	1950–4	1955–9	1960–4	1965–9	1970–4
15–19	.2	.2	0	.2	.2	.2	.2	.2	.2
20–4	.2	.2	0	.2	.2	.2	.2	.2	.2
25–9	.2	.2	0	.2	.2	.2	.2	.2	.2
30–4	.2	.2	0	.2	.2	.2	.2	.2	.2
35–9	.2	.2	0	.2	.2	.2	.2	.2	.2
40–4	0	0	0	.2	.2	.2	.2	.2	0
Period TFR	5.0	5.0	0	6.0	6.0	6.0	6.0	6.0	5.0

Cohort fertility is the fertility (the number of live births) produced by a particular group of women throughout their reproductive life i.e. a real cohort.

Preston, S., Heuveline, P., & Guillot, M. (2000). *Demography: measuring and modeling population processes*.

PERIOD AND COHORT- TEMPO AND QUANTUM

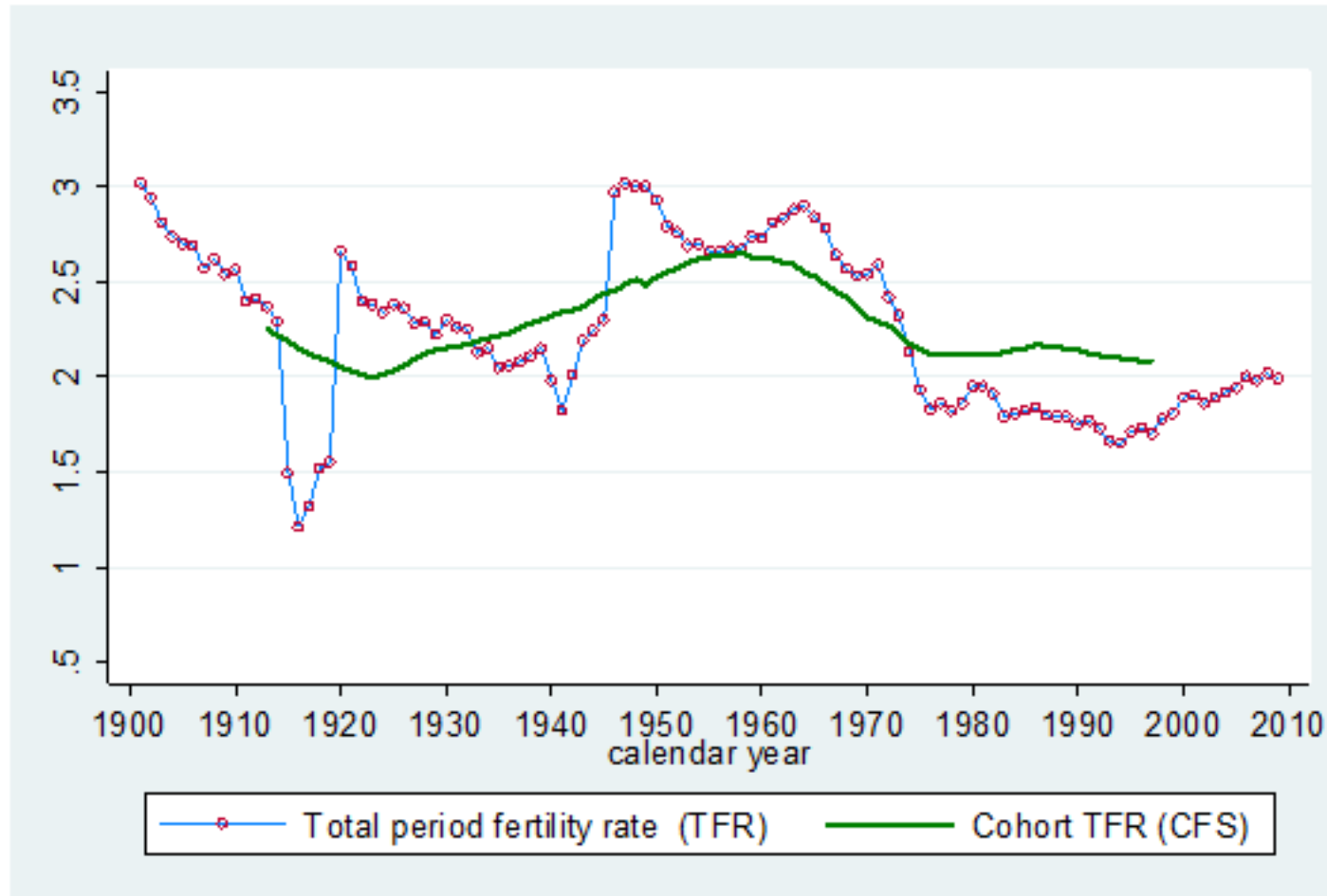


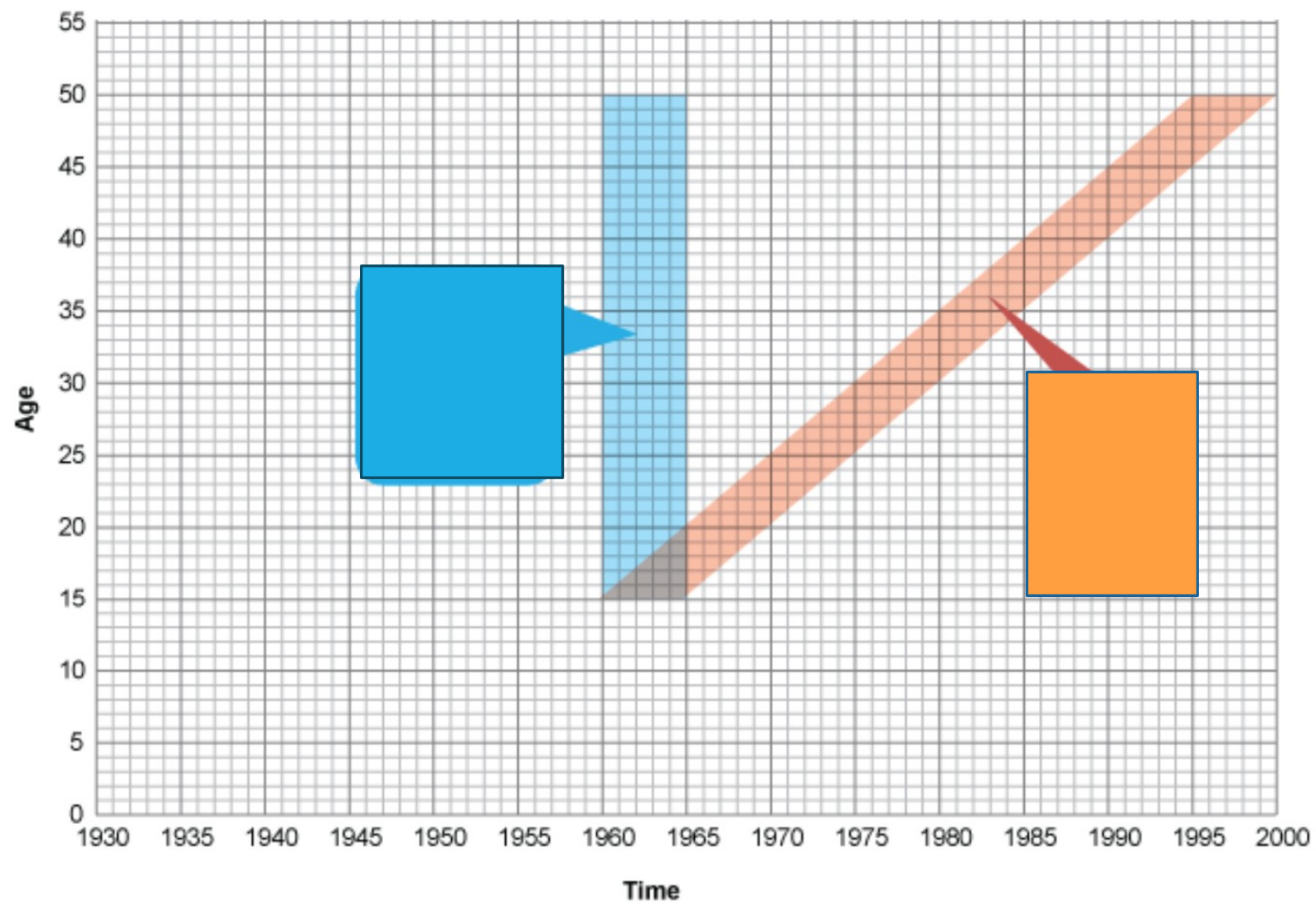
FRANC
E

Tempo: timing of births that has produced an apparent and temporary change.

Quantum relates to- the real level of fertility over a substantial time.

FRANCE – PERIOD AND COHORT





CLASS EXERCISE- PERIOD

Table 2: Age-specific fertility rates, England and Wales, by calendar period.

Age group	1960 - 64	1965 - 69	1970 - 74	1975 - 79	1980 - 84	1985- 89	1990 - 94
15-19	0.040	0.049	0.045	0.030	0.030	0.031	0.033
20-24	0.175	0.165	0.130	0.108	0.095	0.094	0.088
25-29	0.183	0.164	0.135	0.125	0.127	0.125	0.120
30-34	0.105	0.090	0.067	0.065	0.075	0.084	0.087
35-39	0.049	0.040	0.025	0.020	0.024	0.030	0.033
40-44	0.015	0.010	0.007	0.005	0.005	0.005	0.006
45-49	0.000	0.000	0.000	0.001	0.001	0.001	0.001
TFR	2.84	2.59	2.05				

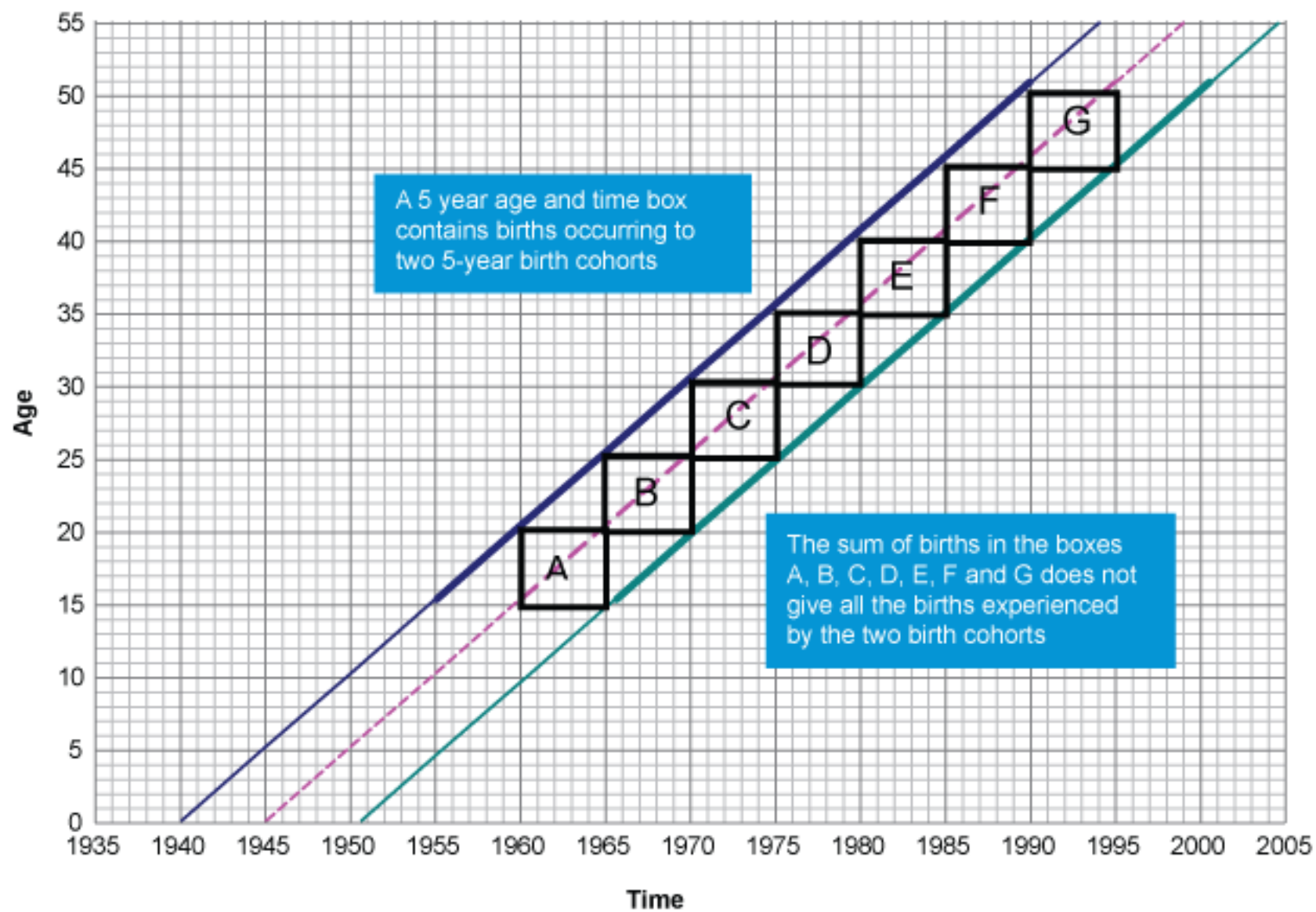
Population Analysis for Policies & Programmes <http://papp.iussp.org/>, Module 101-Session 04

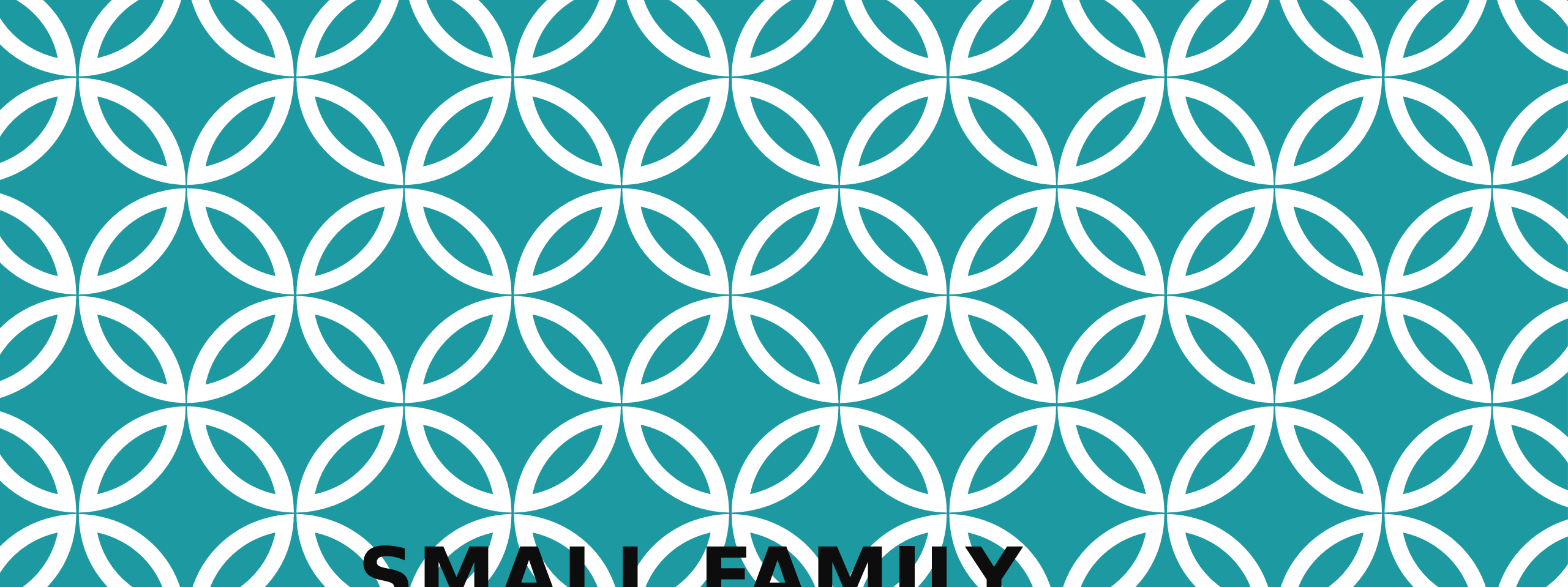
1.77 1.79 1.85 and 1.84

COHORT

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45-49	0.000	0.000	0.000	0.001	0.001	0.001	0.001
TFR	2.84	2.59	2.05				

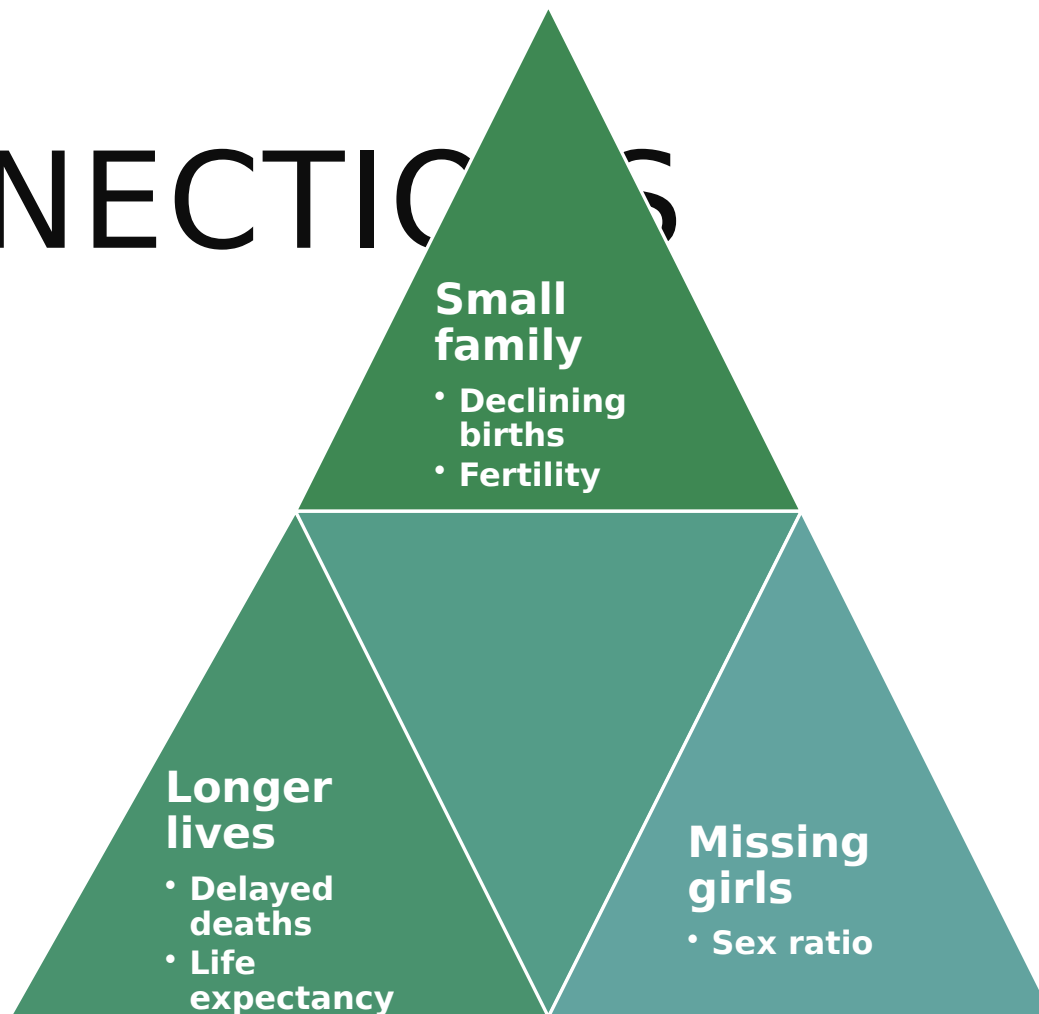




SMALL FAMILY, LONGER LIVES AND MISSING GIRLS: SON PREFERENCE IN INDIA

Adopted from – work of
Mayur Trivedi

CONNECTIONS



Jayachandran, S. (2017). "Fertility decline and missing women." *American Economic Journal: Applied Economics*, 9(1), 118-139.

Ebenstein, A. (2013). "Why do Parents Abort Girls? Patrilocality and Its Historical Origins". Available at: http://www.princeton.edu/rpds/events_archive/repository/Ebenstein021214/Ebenstein021214b.pdf

DEMOGRAPHIC PROCESSES

Birth

• Fertility

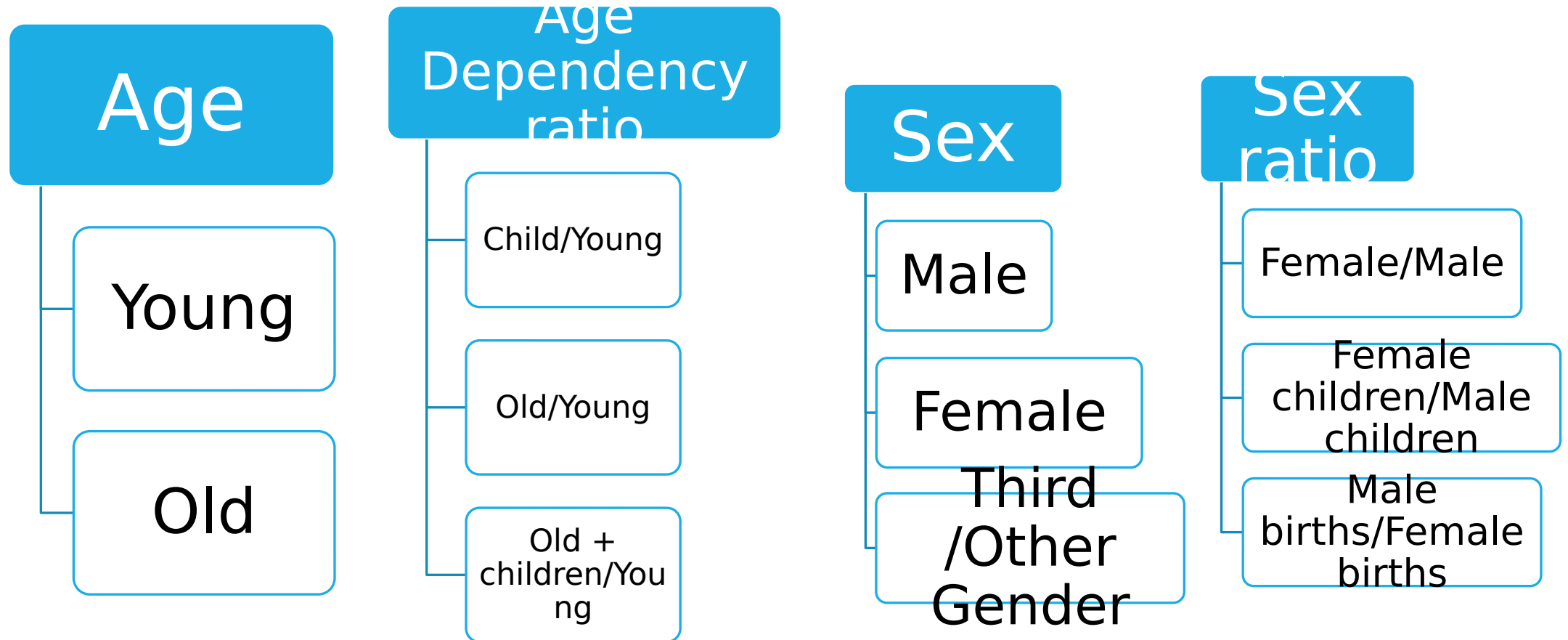
Death

• Mortality

Movement

• Migration

COMPOSITION



TOTAL FERTILITY RATE

Most commonly used fertility indicator

Number of children a woman will have if she lives through all the reproductive ages and follows the age-specific fertility rates of a given time period

Clear reflection of average no. of children born to a women

- Higher the TFR, More children per women, bigger the average family size
- Lesser the TFR, smaller families

SEX RATIO

All age sex ratio

- War, Single male migration etc.

Sex ratio at birth

- Interventions at or before birth

Child sex ratio

- Unwanted girl fetuses--unwanted children
- "Extended infanticide" : be deprived of nutrition and medical care, thus being more vulnerable to infant and child mortality

POPULATION AGING

Population change affects size, structure and composition of population

- Rapid growth – more children than adults and elders
- **Slow growth – less children than adults and elders**

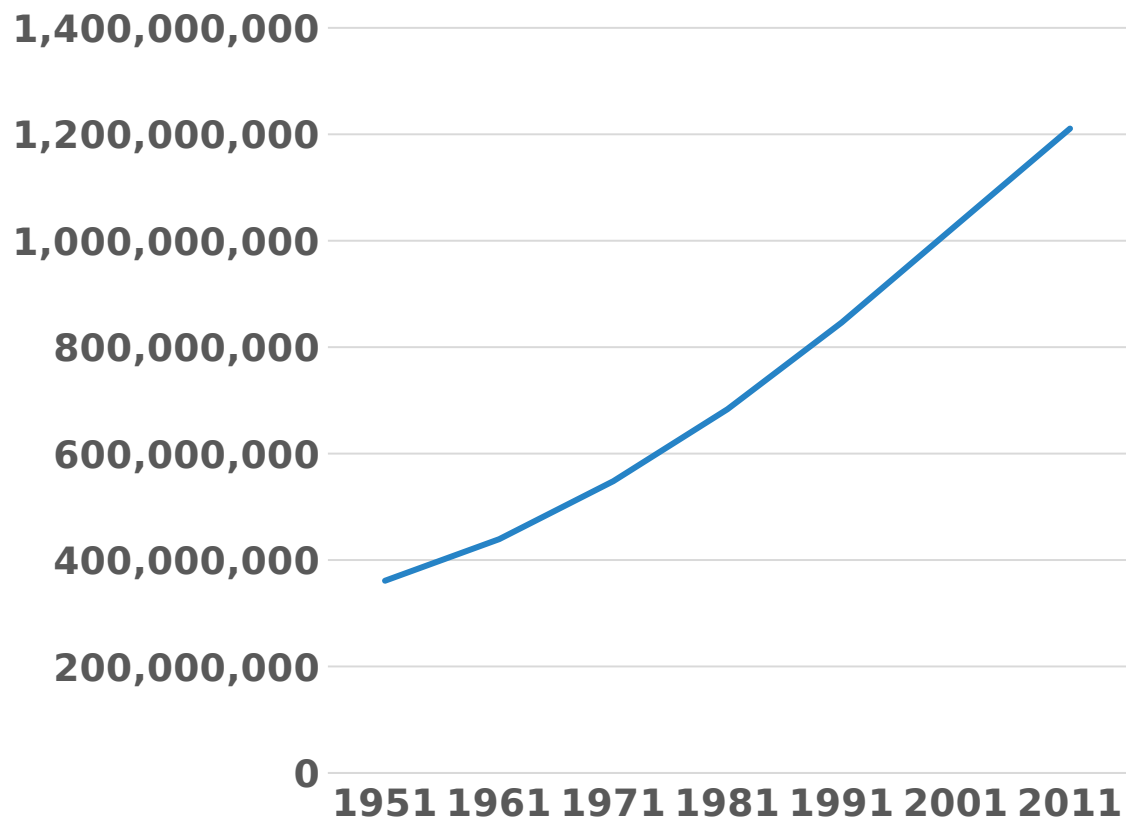
In rapid growth demography, early marriage and childbirth is common; quick replacement of workforce

In slow growth demography, proportion of elderly (aged more than 60 years) increases rapidly

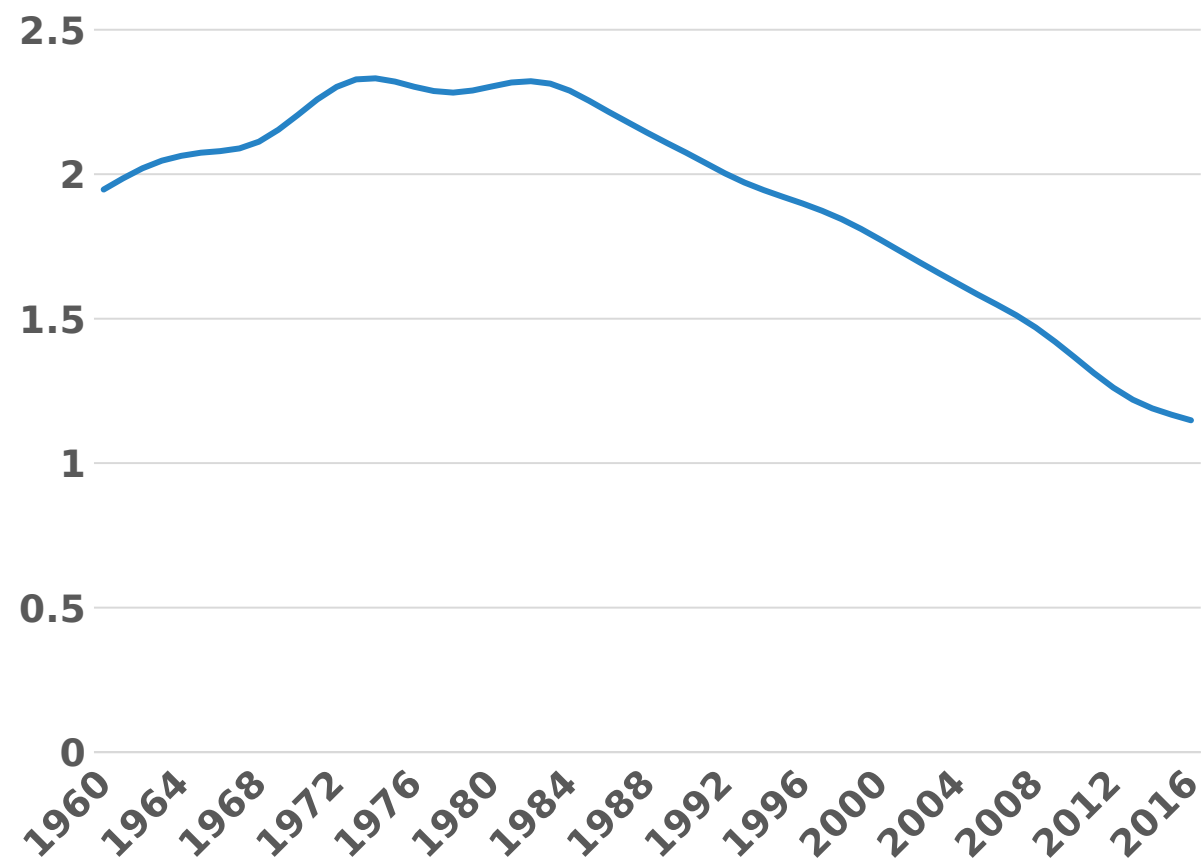
- Old age care increasingly becomes an important need in a household as well as in a society

POPULATION IN INDIA

Population



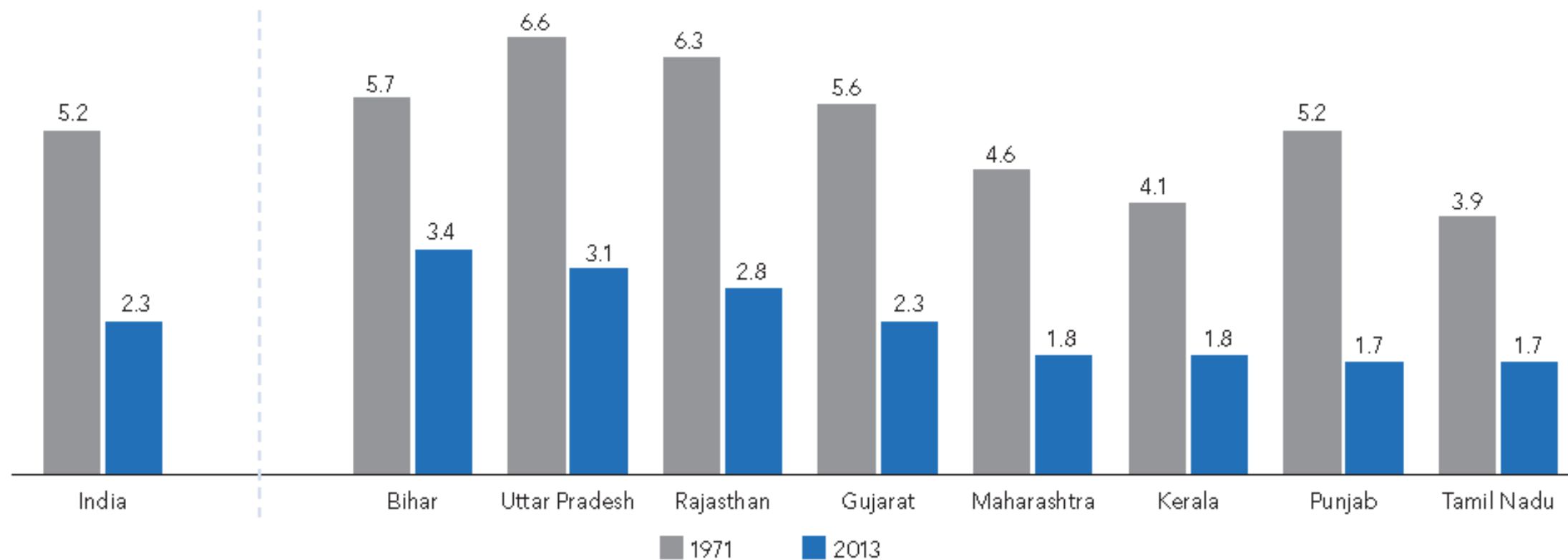
Population growth rate (annual %)



INDIA TFR

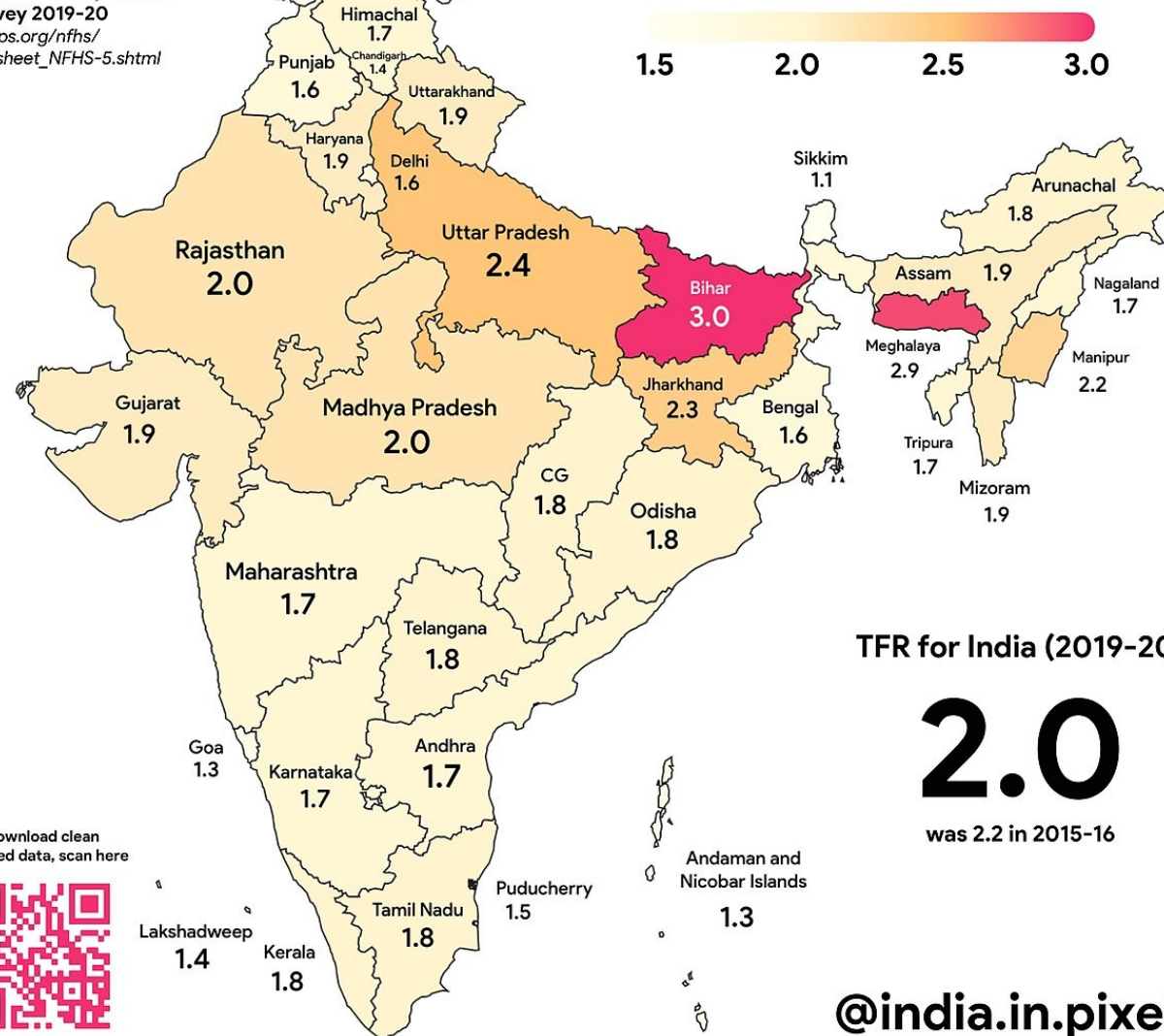
Total Fertility Rate, India and Selected States, 1971 and 2013

Children Per Woman



Source:
National Family Health
Survey 2019-20
[rchiips.org/nfhs/
factsheet_NFHS-5.shtml](https://rchiips.org/nfhs/factsheet_NFHS-5.shtml)

Total Fertility Rate for all States (2019-20)



TFR for India (2019-20)

2.0

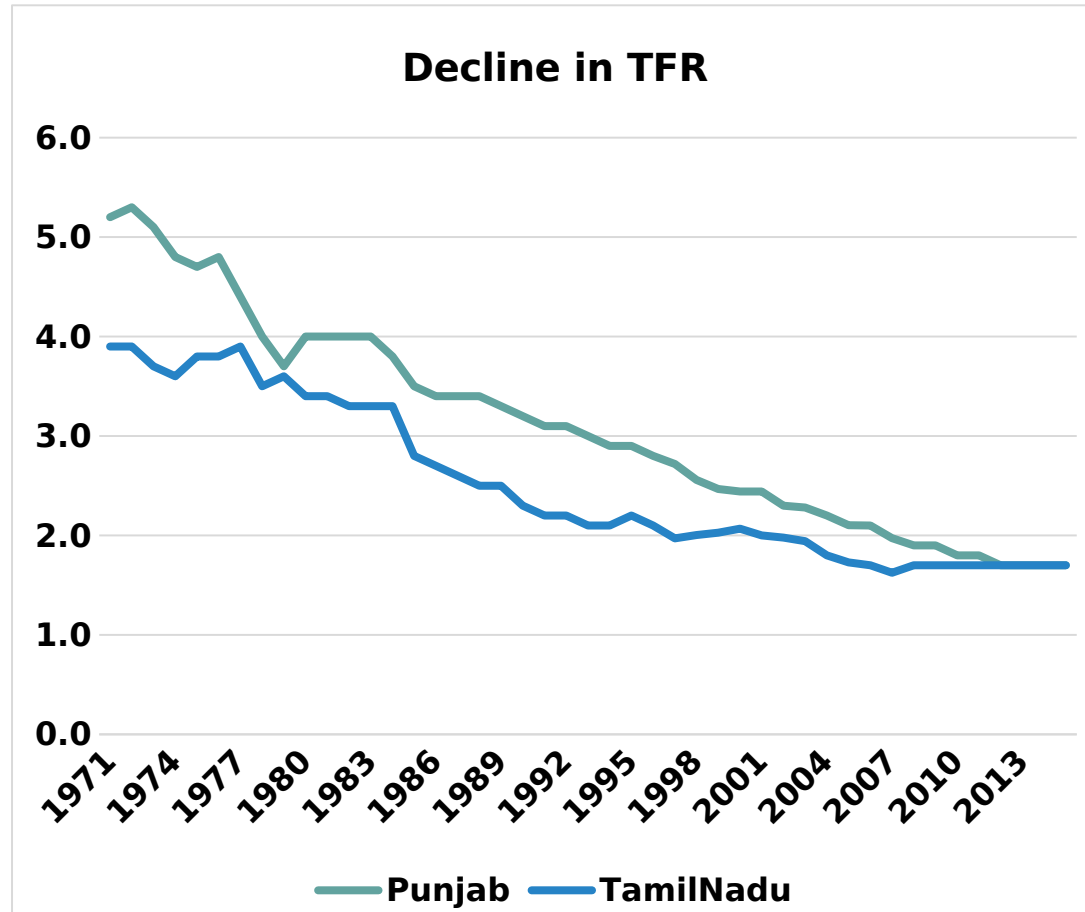
was 2.2 in 2015-16

To download clean
parsed data, scan here



@india.in.pixels

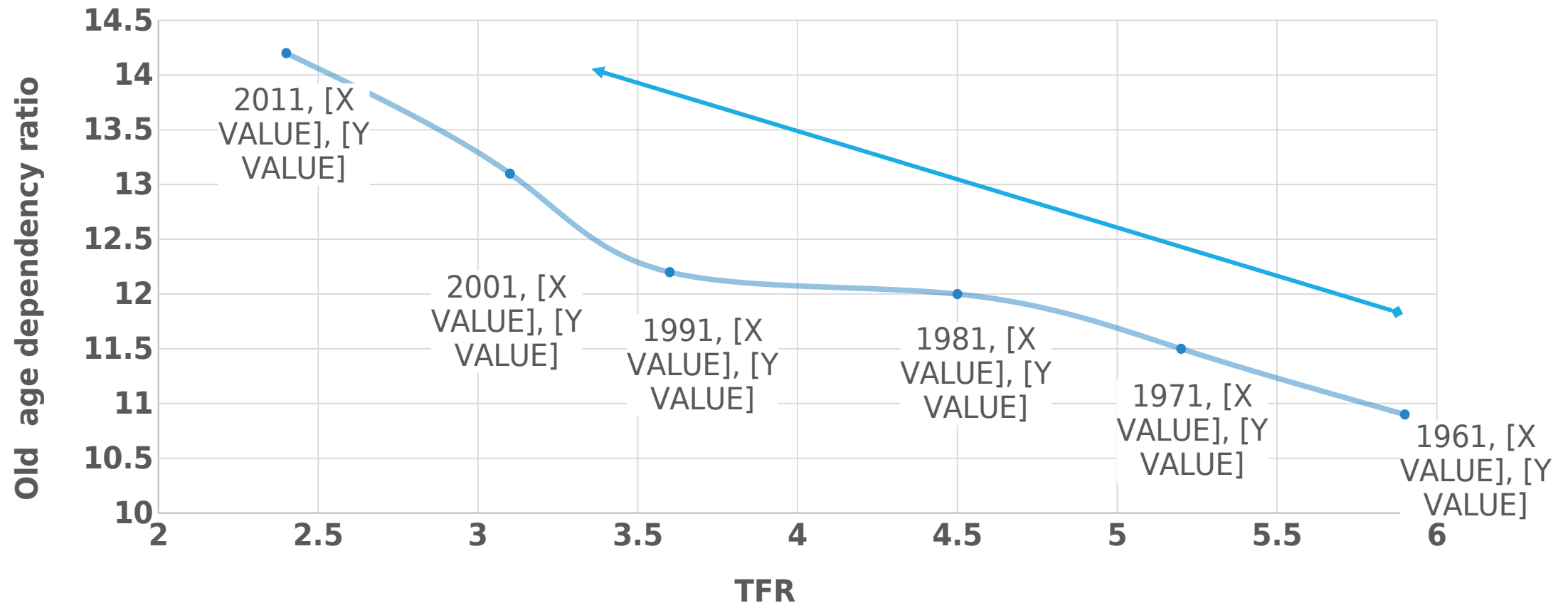
TFR TREND IN INDIA



Punjab and Tamil Nadu have TFR of 1.7 in 2013

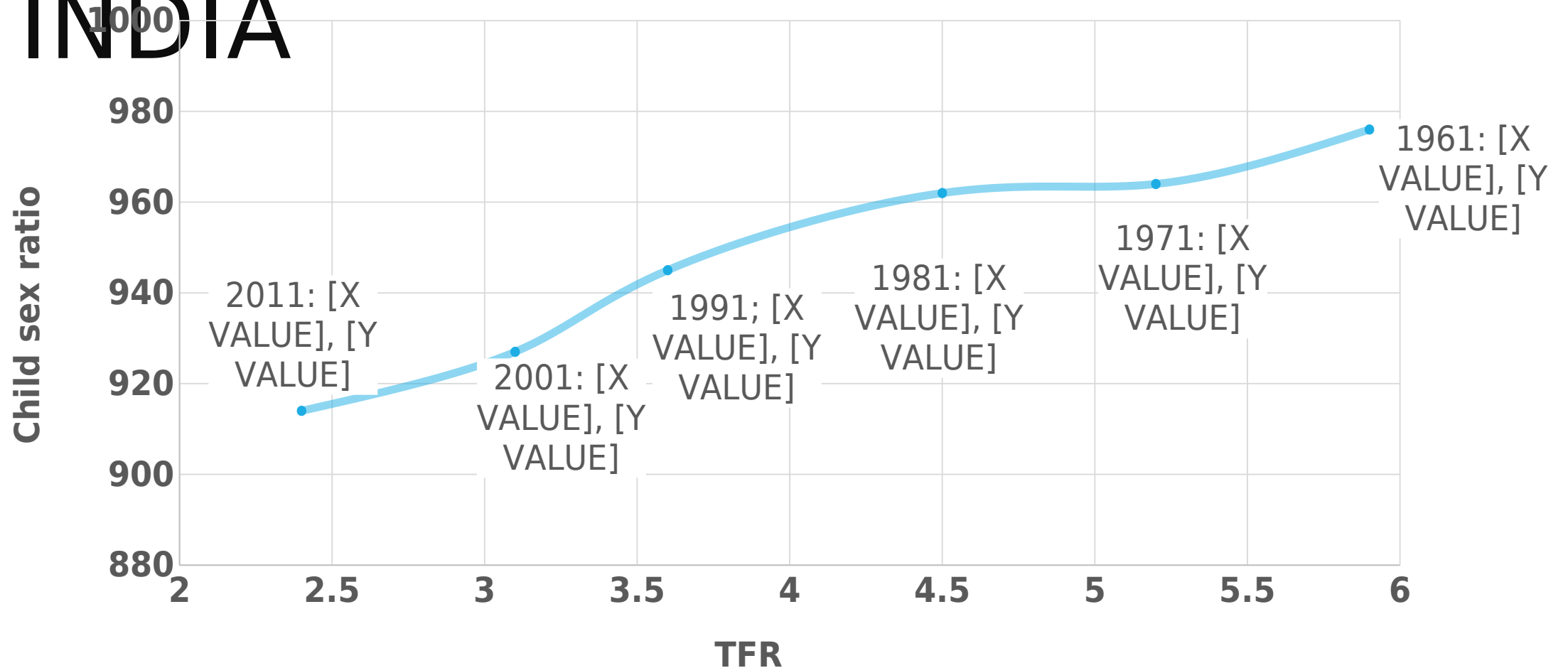
- However, their journey have been different
- Punjab seen sharp and rapid decline in fertility; TN saw gradual and slow decline
- Rapid decline in fertility affect demographic decisions in quick successive generations

OLD AGE DEPENDENCY AND TFR - INDIA



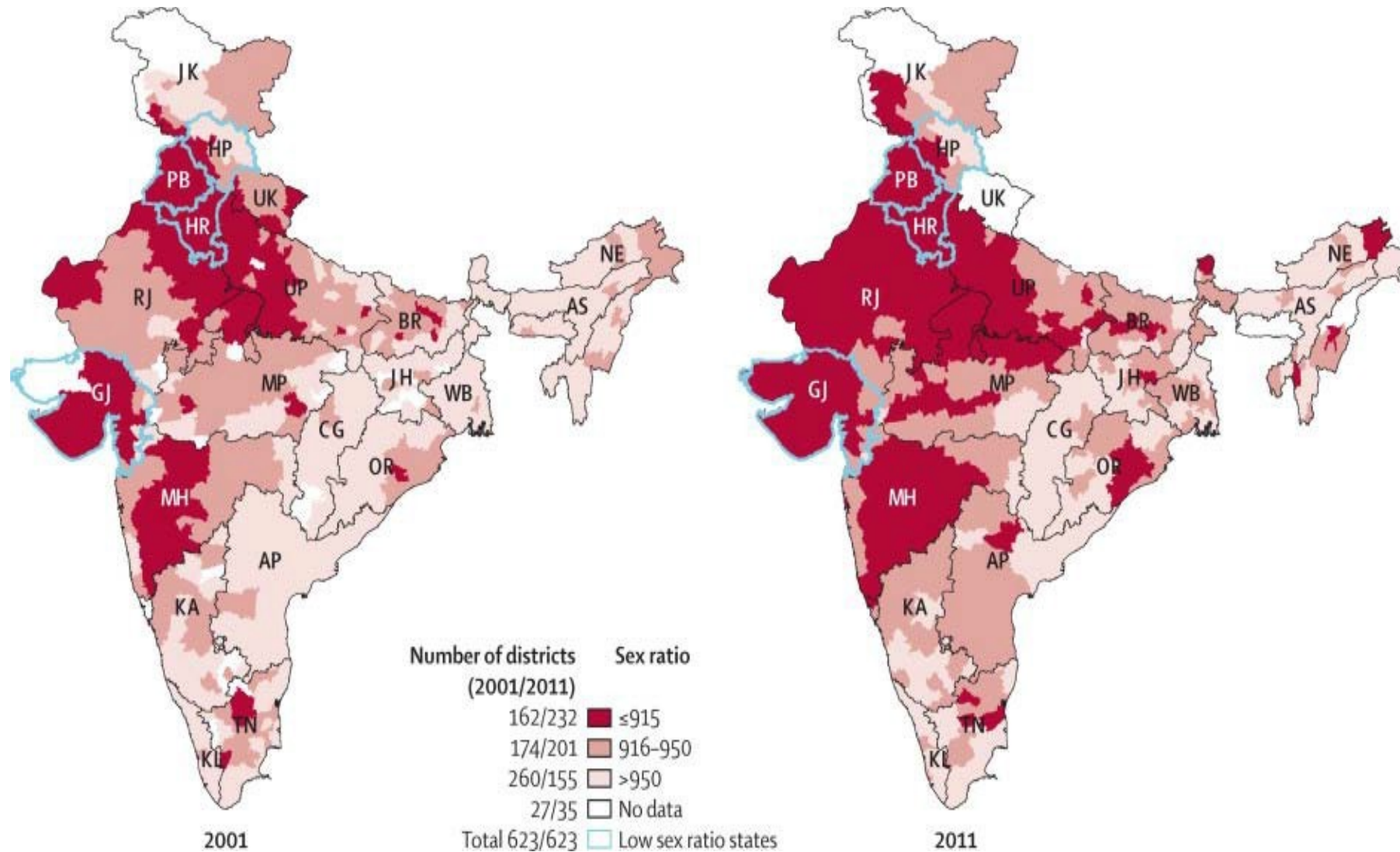
Source: Various censuses, Compendium of India's Fertility and Mortality Indicators, 1971 - 2013, and elderlies in India 2016

CHILD SEX RATIO AND TFR - INDIA



Source: Various censuses, Compendium of India's Fertility and Mortality Indicators ,1971 - 2013, and world development indicators

Child sex ratio change 2001- 2011

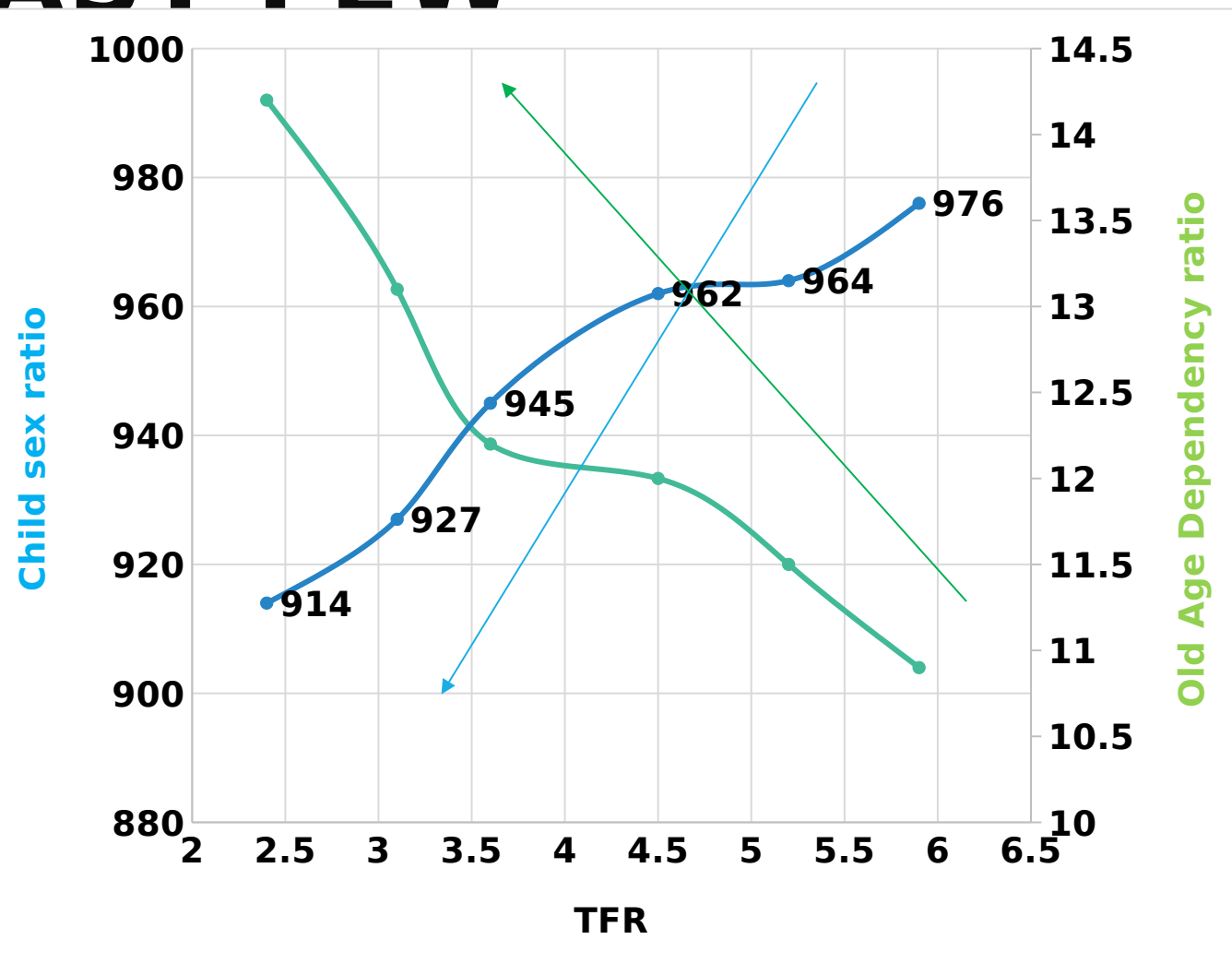


CONNECTIONS OF THREE CHANGES IN LAST FEW DECADES

Fertility
declined

Child sex ratio
worsened

old age
dependency
increased



WHAT CAN AFFECT SEX RATIO?

Technology

- Amniocentesis and ultra-sonography

Statutes/legislations

- Collusion between people, the medical fraternity and the administration

Son preference

- Lower demand for daughters

WHY SON PREFERENCE?

Declining fertility and increasing population aging !!!

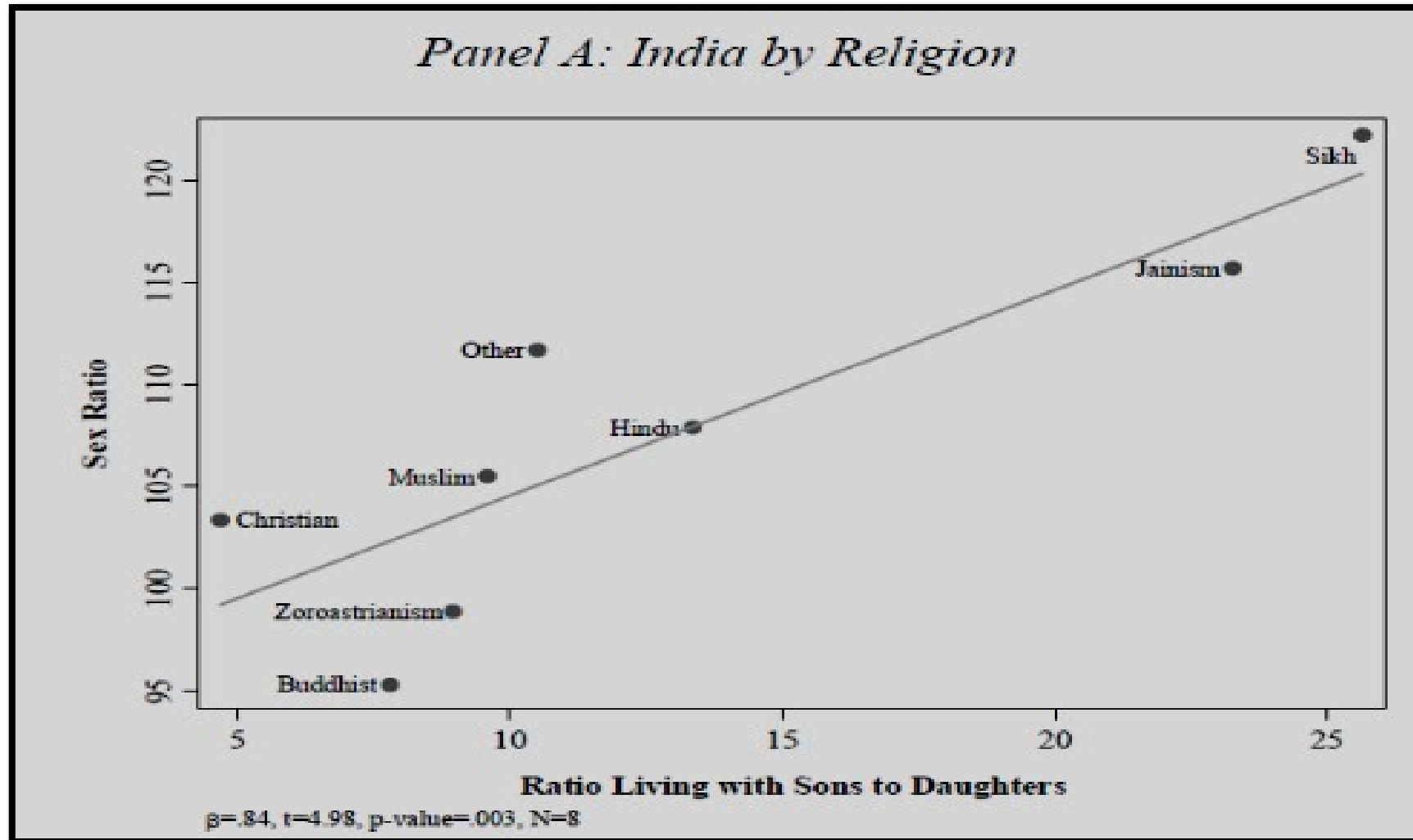
Employment and social security in old age

- 92% of total employment (2012-13; NSSO) are in informal sector; most of these lack pension
- Family remains conventional source of old age social and

Patrilocality

- A cultural norm in which sons provide care for their elderly parents, and daughters leave the home following marriage to provide care for their in-laws
- Common to the social norms of Christians in Armenia, Muslims in Azerbaijan, Hindus in India and Buddhists in China

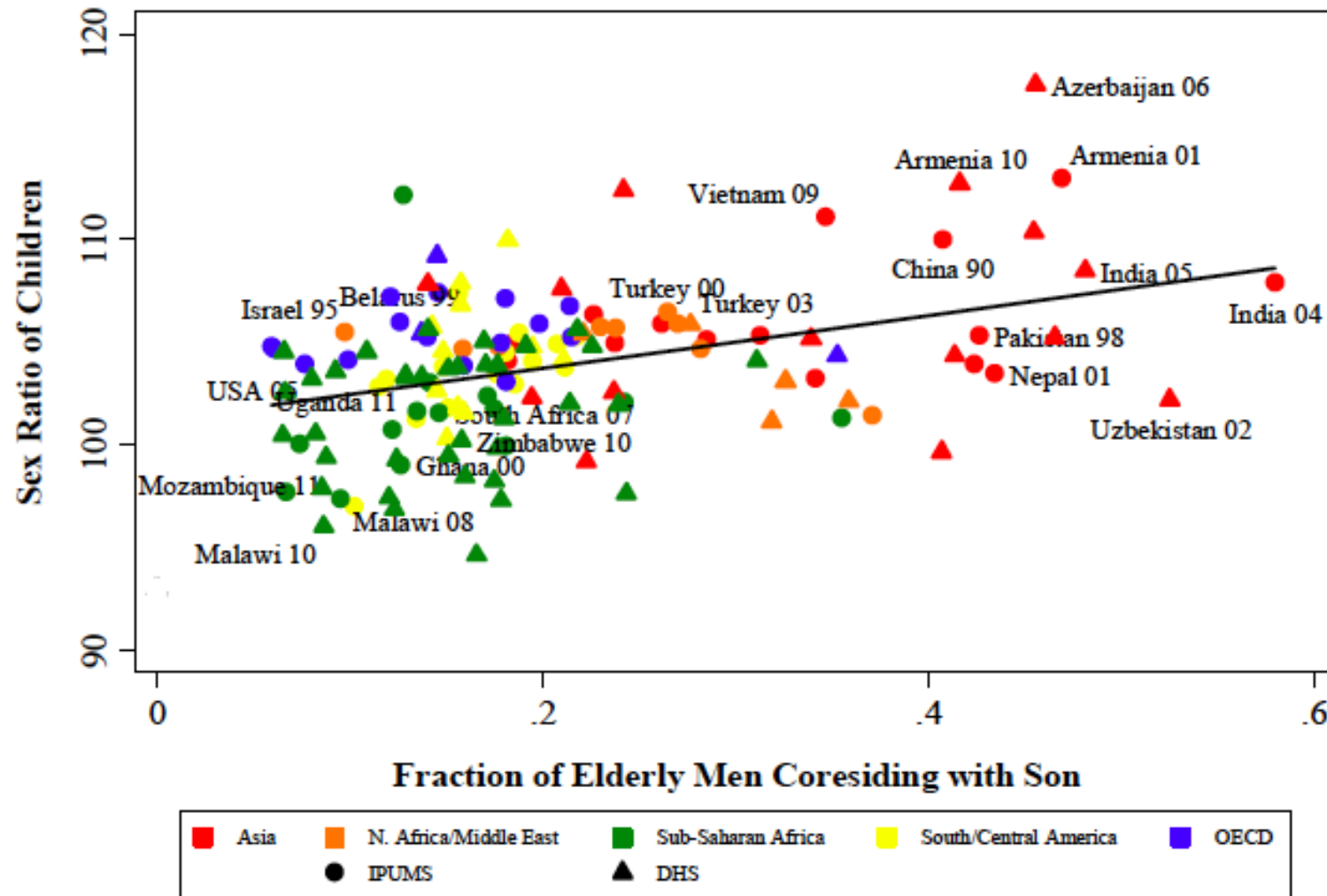
CONNECTION OF CO-RESIDENCE AND SEX RATIO



Source: Avraham Ebenstein 2013 (Why do Parents Abort Girls? Patrilocality and Its

Figure I

Coreidence and the Sex Ratio of Children, IPUMS and DHS Samples



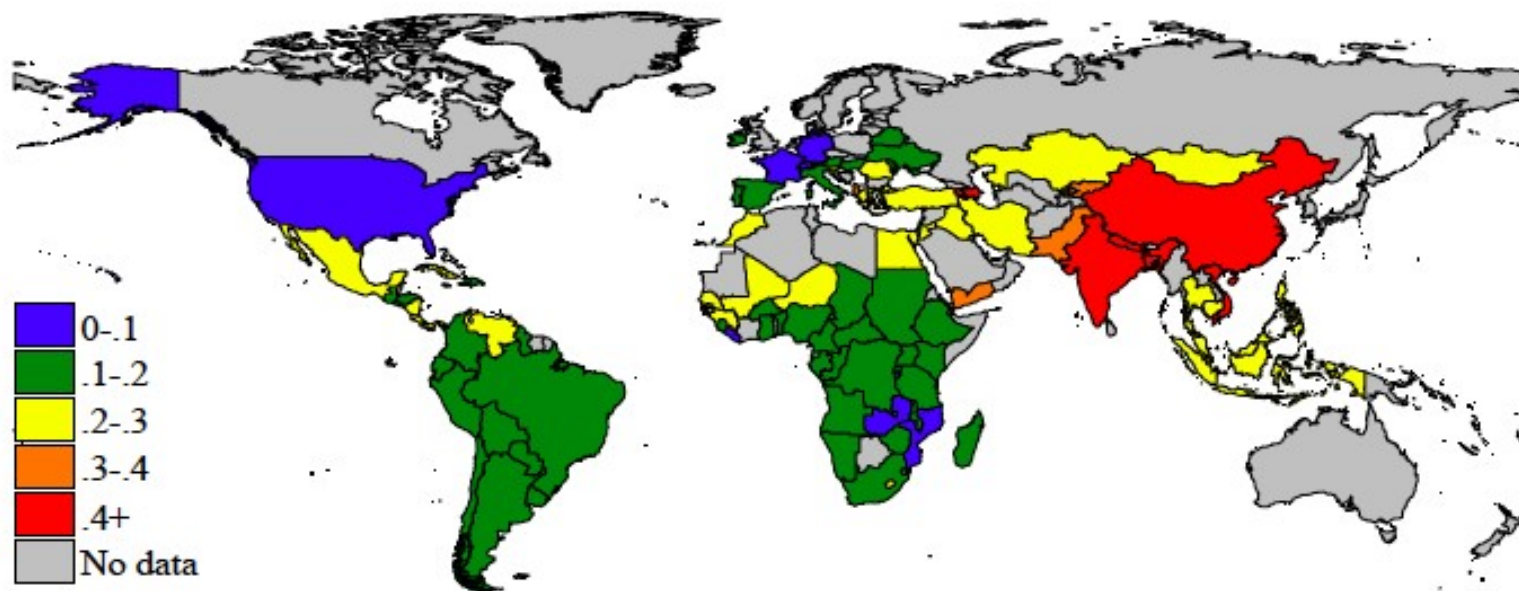
$\beta=12.88$, $t=4.03$, $p\text{-value}<.001$, $N=140$

Notes: The x-axis is the fraction of elderly men (ages 60+) living with their adult son (ages 25+) in the most recent IPUMS and DHS survey for each country. The y-axis is the sex ratio at birth for children aged 0 to 5 in each survey. The sample is composed of 140 surveys from 107 countries.

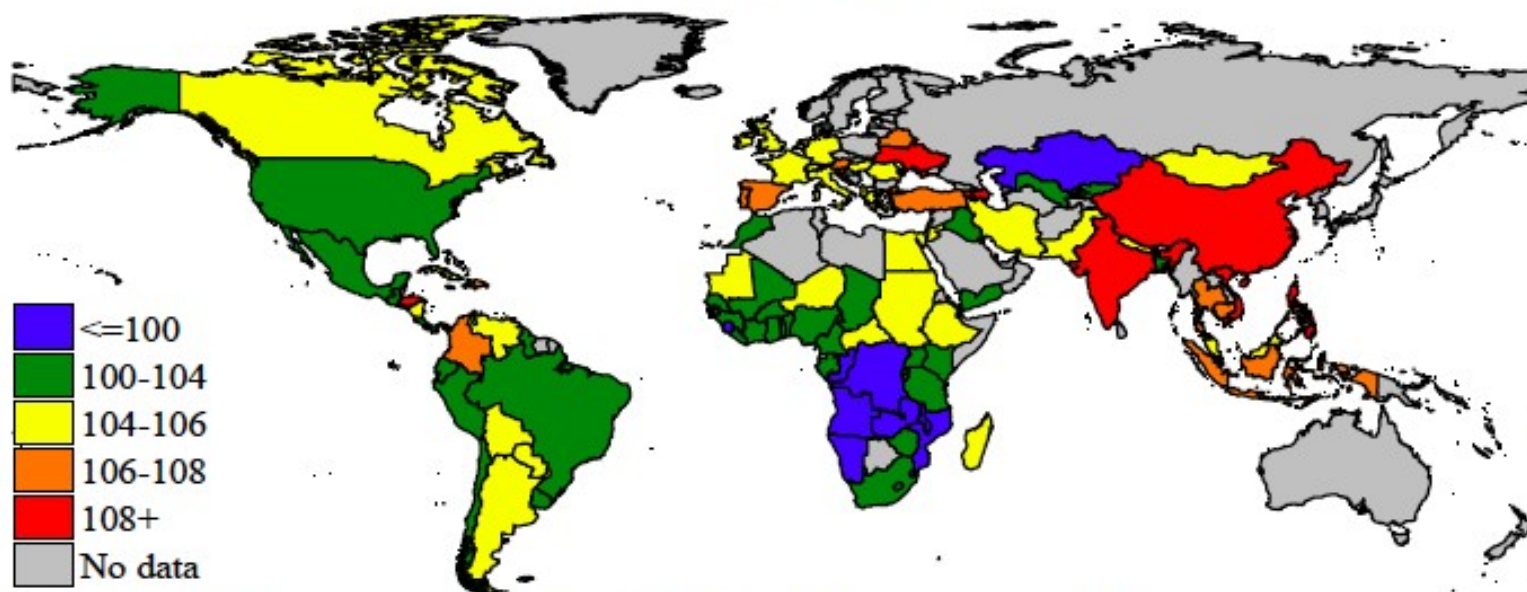
Ebenstein, A. (2013).
Why do Parents Abort
Girls? Patrilocality and
Its Historical Origins.

Figure II

Panel A: Coresidence Rates by Country



Panel B: Sex Ratios by Country



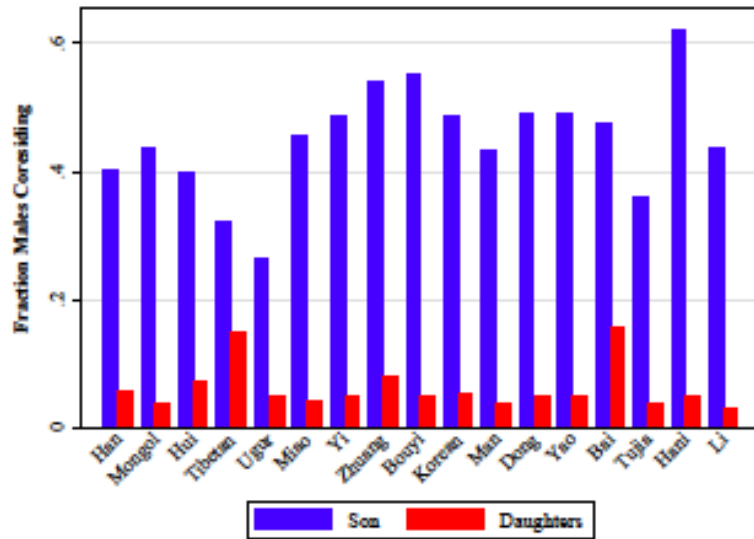
Notes : Data for 103 countries using IPUMS and DHS data with the most recent available survey.

Ebenstein, A. (2013). Why do Parents Abort Girls? Patrilocality and Its Historical Origins.

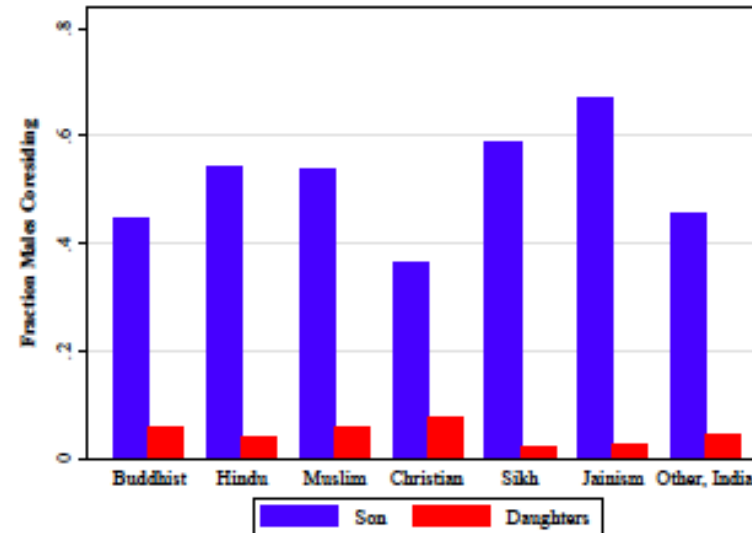
Figure IV

Elderly Coresidence Rates with Sons and Daughters in China, India, and the United States

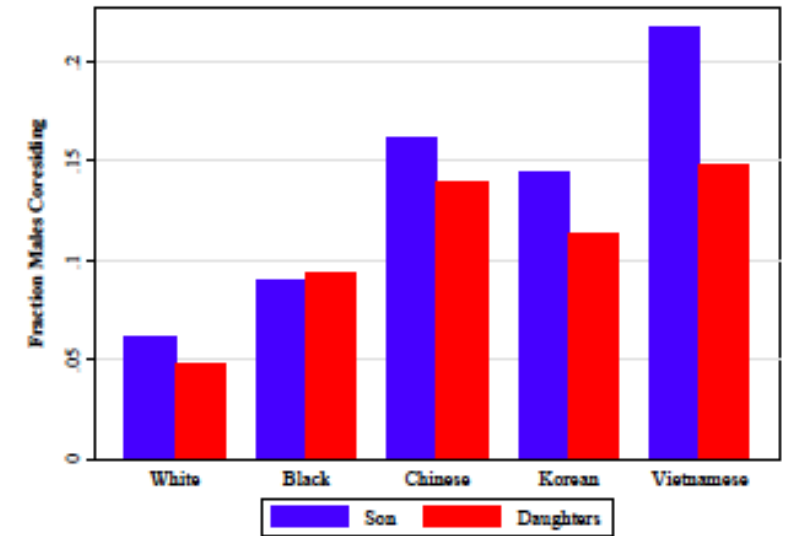
Panel A: China Ethnicities



Panel B: India Religions



Panel C: United States Ethnicities



Notes : China 1990 IPUMS (10% sample), India 2005/6 DHS Sample, United States IPUMS 2010. See Figure 1 for details.

Ebenstein, A. (2013). Why do Parents Abort Girls? Patrilocality and Its Historical Origins.

FERTILITY, SON PREFERENCE, AND CHILD SEX RATIO

In societies, where four or six children were common, son preference did not need to exist at the expense of daughters

As fertility declines, son preference becomes apparent

With couples wanting two/one children (or, as in China, are allowed only one), they will sacrifice unborn daughters to their pursuit of a son

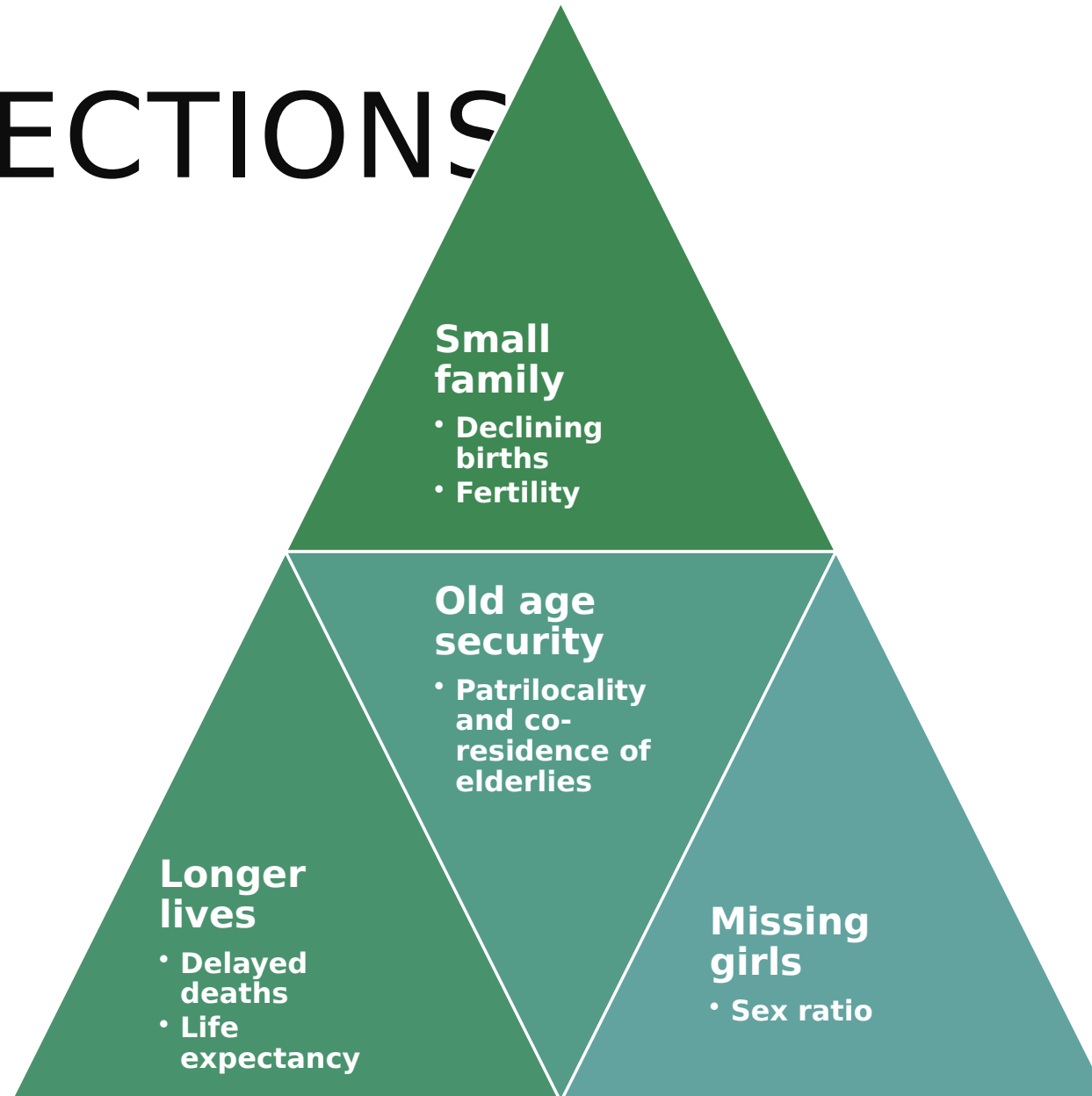
OLD-AGE CARE AND SEX-RATIO

The “missing girls” phenomenon is related less to religion or culture and more to the role played by sons in caring for their elderly parents.

While initiatives aimed at improving female status are noble, expanding social insurance is likely to be more effective at normalizing the sex ratio.

- Avraham Ebenstein 2013 -Why do Parents Abort Girls? Patrilocality and Its Historical Origins

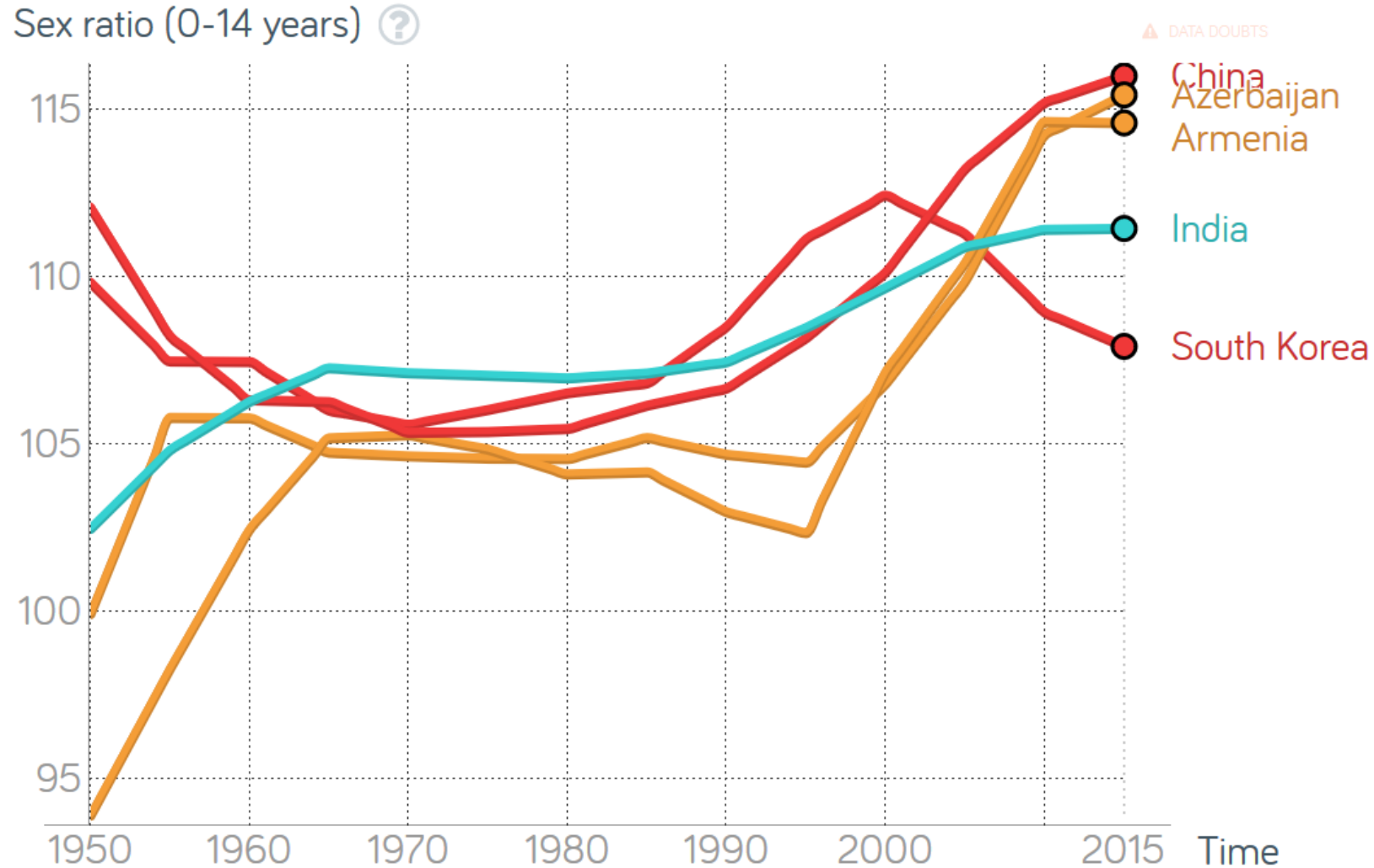
CONNECTIONS



ARE WE DOING ENOUGH?

- PNDDT ACT is necessary but not sufficient
 - In the possible absence of the PNDDT Act, child sex ratio would have declined by another 13-20 points, or an additional 51,000 female fetuses would have been aborted. (Nandi and Deolalikar 2011)
- Causes of causes
 - Girl child vs. status of women
- Patriarchy and Patrilocality
 - Social revolution?!!!
 - Social security revolution?!!

SOUTH KOREA INTRODUCED PENSION SCHEME IN 1995!



SUMMARY

Household decisions regarding births that are rooted in socio-economic realities have long term impact at societal level.

Declining child sex ratio in India needs to be seen contextually

Ban on sex selection can have low to modest effect in reversing the trend

For long term effect, efforts towards gender equity are needed

It is important to understand the root-causes of son preference and address them through structural and policy changes, and sustained advocacy

REFERENCES

Sen, A. (1992): "Missing Women," British Medical Journal, 304, 586-87

Jayachandran, S. (2017). "Fertility decline and missing women." *American Economic Journal: Applied Economics*, 9(1), 118-139.

Ebenstein, A. (2013). "Why do Parents Abort Girls? Patrilocality and Its Historical Origins". Available at:
http://www.princeton.edu/rpds/events_archive/repository/Ebenstein021214/Ebenstein021214b.pdf

Das Gupta, Monica, "Missing Girls In the South Caucasus Countries: Trends, Possible Causes, and Policy Options" (April 9, 2015). World Bank Policy Research Working Paper No. 7236. Available at SSRN: <https://ssrn.com/abstract=2592781>



China sees lowest number of births amid concerns over population decline

1 min read • 12 Oct 2023, 02:53 PM IST

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News > World News > Japan Birth Rate Hit Record Low In 2023 As Nation's Population Crisis Deepen

Japan Birth Rate Hit Record Low In 2023 As Nation's Population Crisis Deepen

The number of births fell 5.1% from a year earlier to 758,631, while the number of marriages slid 5.9% to 489,281 - marking the first time in 90 years the number fell below 500,000 - foreboding a further decline in the population as out-of-wedlock births are rare in Japan.

Pet theme: Editorial on India’s fertility rate and Modi government’s wish for population stability

Join us

Widespread information and accessibility of birth control methods, better education, urbanisation, small families and economic necessity have led to the gradual drop in TFR over decades

The Telegraph

The Editorial Board | Published 07.02.24, 06:51 AM

Goa's population growth stalls, workforce may feel pressure

Gauree Malkarnekar / TNN / Updated: Feb 26, 2024, 09:59 IST

71 PTS

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Goa's natural population growth rate is projected to remain flat at 0.4% annually until 2036. The birth rate has fallen during the pandemic, and an unusually high number of deaths were reported in Goa in 2021 due to the Covid pandemic. Goa's li ... [Read More](#)

ACTIVITY

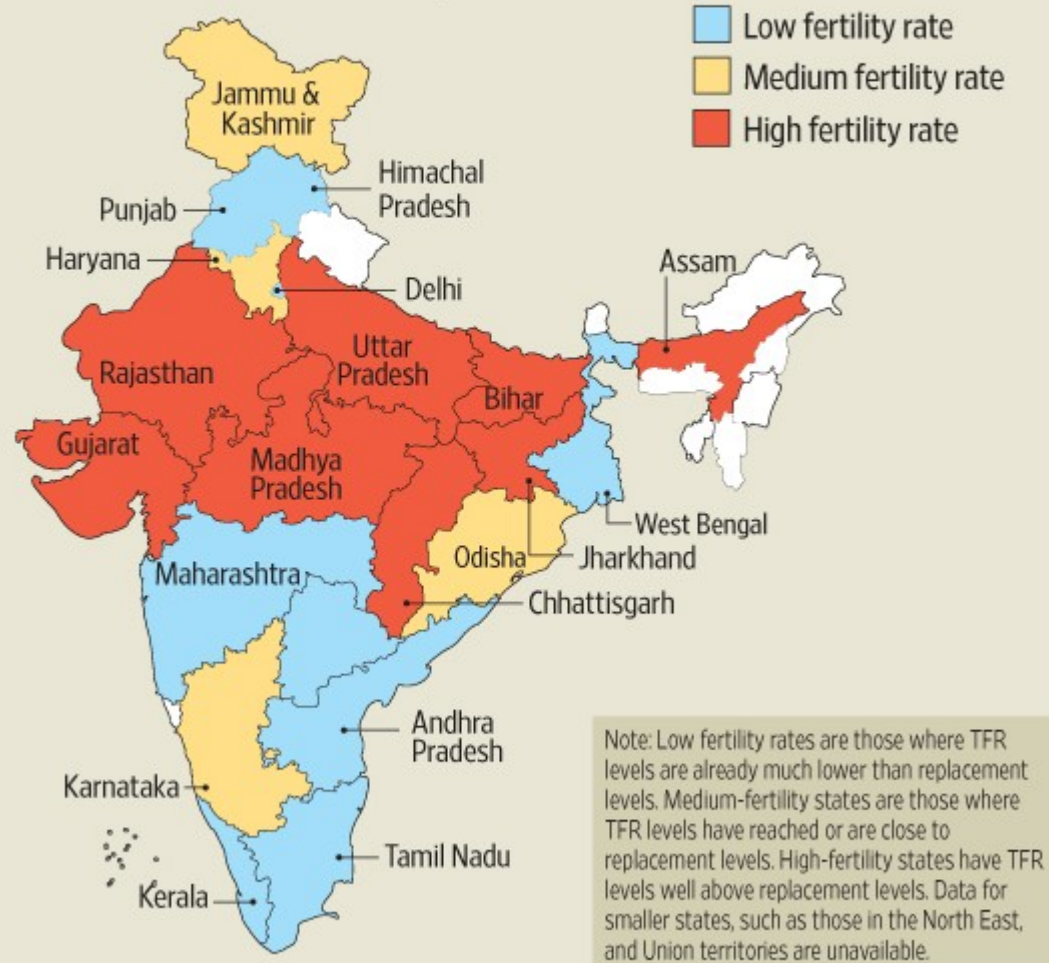
By next class scan through top National/International newspapers and magazines (Indian Express, The Hindu, Scroll, Mint, The Wire,... TOI, New York Times, Wall Street Journal, CNN, BBC...)- Feb-March 2024

1. Identify the contemporary news article on Population – Fertility, Mortality, ... Migration
2. Data Sources and Data Inference from the news article
3. Policies related to the population issue addressed.

2013

FERTILITY LEVELS ACROSS MAJOR INDIAN STATES

Total fertility rates of 10 Indian states are below replacement levels; population in these states is expected to decline in the coming years. In almost all low-fertility states, female literacy rates are above the national average, whereas in almost all high-fertility states, female literacy rates are below the national average of 65%.



State	Total fertility rate	Female literacy rate (%)
West Bengal	1.6	71.16
Delhi	1.7	80.93
Himachal Pradesh	1.7	76.6
Punjab	1.7	71.34
Tamil Nadu	1.7	73.86
Andhra Pradesh	1.8	59.74
Kerala	1.8	91.98
Maharashtra	1.8	75.48
Jammu & Kashmir	1.9	58.01
Karnataka	1.9	68.13
Odisha	2.1	64.36
Haryana	2.2	66.77
Assam	2.3	67.27
Gujarat	2.3	70.73
Chhattisgarh	2.6	60.59
Jharkhand	2.7	56.21
Rajasthan	2.8	52.66
Madhya Pradesh	2.9	60.02
Uttar Pradesh	3.1	59.26
Bihar	3.4	53.33

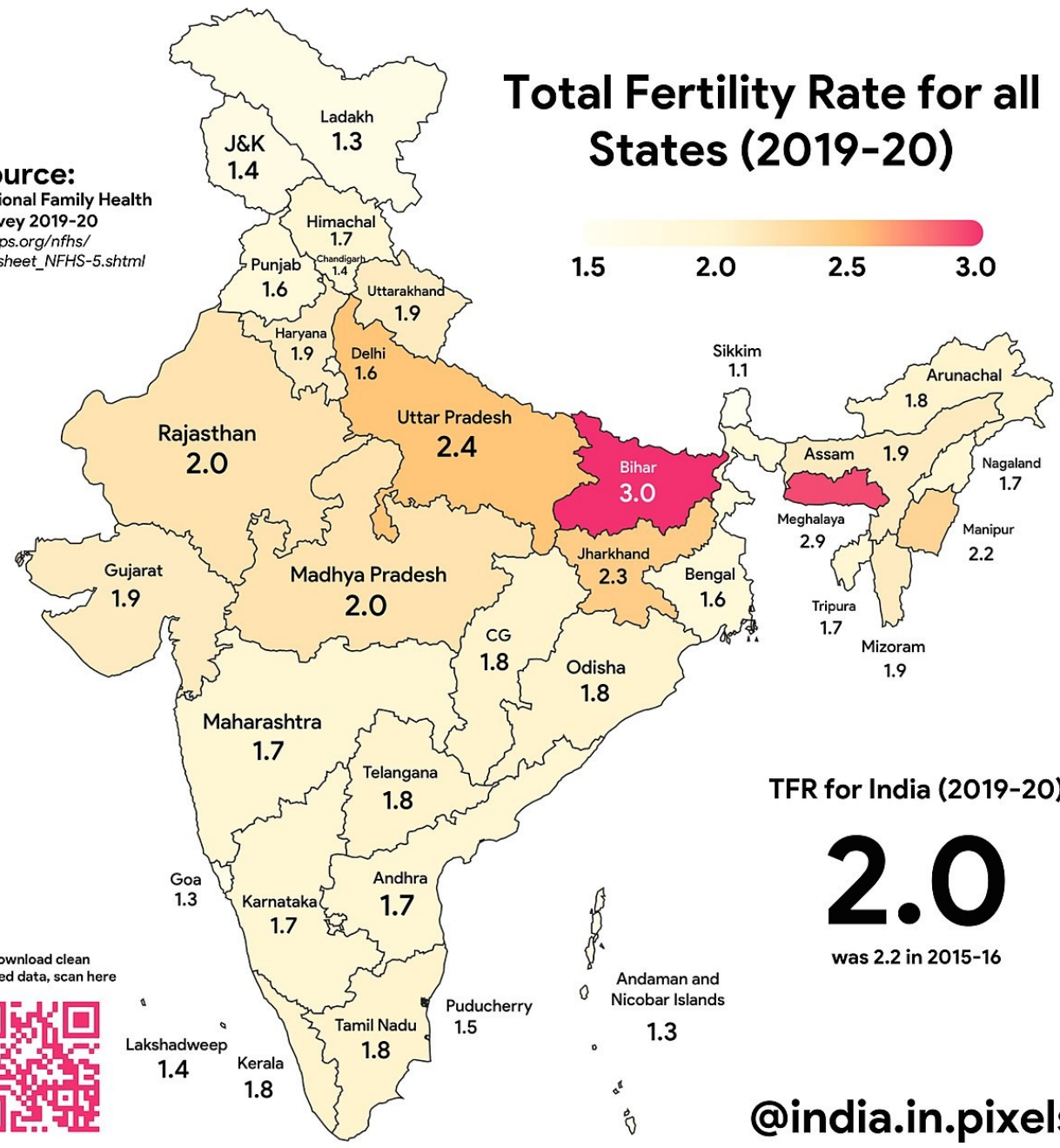
Graphic by Ajay Negi/Mint

Source: SRS

Data: SRS 2013

Why West Bengal is like Canada, and Bihar like Swaziland Fertility rates vary dramatically across the country, Premit Bhattacharya, Livemint Thu, Dec 25 2014

Source:
National Family Health
Survey 2019-20
[rchiips.org/nfhs/
factsheet_NFHS-5.shtml](https://rchiips.org/nfhs/factsheet_NFHS-5.shtml)



TFR –
NFHS-5
2019-20

TFR for India (2019-20)

2.0
was 2.2 in 2015-16

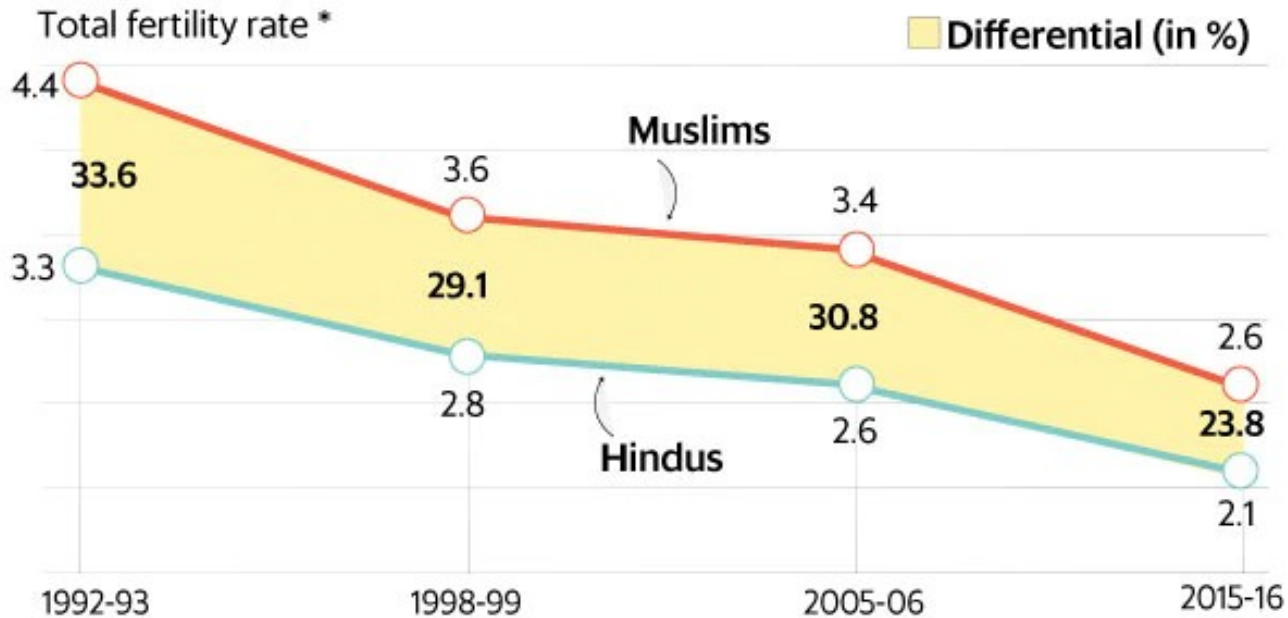
To download clean
parsed data, scan here



@india.in.pixels

Chart 1

After staying around 30% since 1992-93, the Hindu-Muslim fertility differential narrowed to 23.8% in 2015-16



* Average number of children a woman will have in her lifetime at current fertility levels

Source: National Family Health Survey (NFHS)

<https://www.livemint.com/news/india/what-a-narrowing-hindu-muslim-fertility-gap-tells-us-1550686404387.html>

live**mint**

NFHS-5 data: Total fertility rate dips, sharpest decline among Muslims

In NFHS 5, the country's overall fertility fell below the replacement level of two children per woman, falling from 2.2 in NFHS 4.

Esha Roy , May 10, 2022

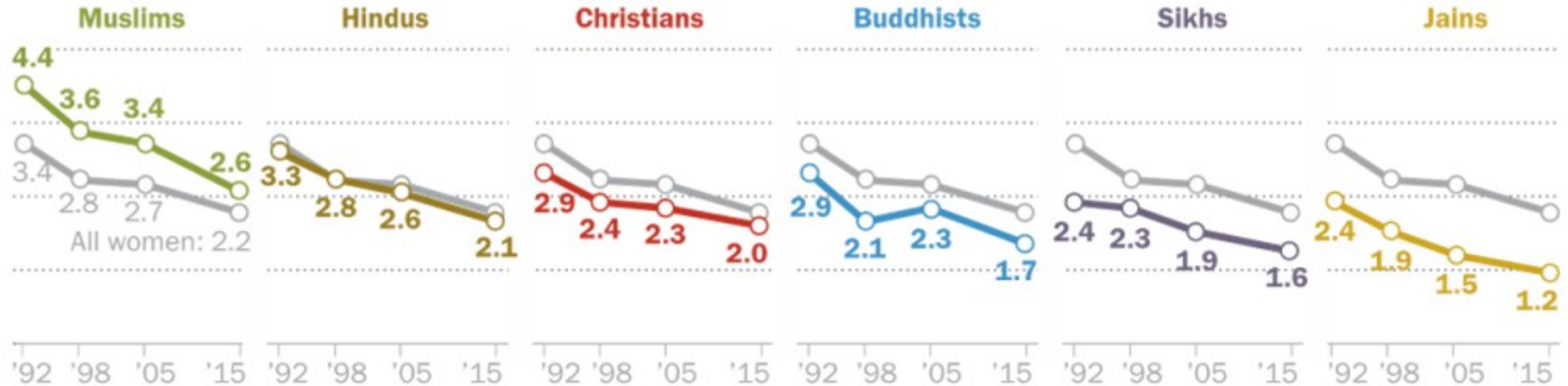
“In keeping with the downward trend seen over the years, the community's fertility rate dipped to 2.3 in 2019-2021 from 2.6 in 2015-16. While all religious communities have shown a decline in fertility, contributing to a dip in the nation's total fertility rate, the fall has been sharpest in the Muslim community, from 4.4 in NFHS 1(1992-93) to 2.3 in NFHS 5(2019-2021).

In NFHS 5, the country's overall fertility fell below the replacement level of two children per woman, falling from 2.2 in NFHS 4.”

 **The Indian EXPRESS**

In India, fertility rates have fallen and religious gaps have shrunk

Number of children an average woman in India is expected to have in her lifetime



Source: National Family Health Survey, 1992-2015.
"Religious Composition of India"

PEW RESEARCH CENTER

https://www.pewforum.org/2021/09/21/religious-composition-of-india/pf_09-21_india_demography_0-1/

Why do we need to educate people about the population?



Shivakumar Jolad

12 min read · Jun 23, 2023



False notions about the population have created fear, prejudice, and communal hatred.

Shivakumar Jolad and Trishala S

<https://medium.com/@shiva-jolad-46811/why-do-we-need-to-educate-people-about-the-population-5ca41e1e43fc>

DEMOGRAPHIC DATA SOURCES AND COLLECTION

Demographic Data: Sources, Collection and Evaluation

- ▢ Censuses and vital registration
- ▢ **Single-round surveys**
- ▢ Longitudinal studies